

Invited review

The pharmacology of voltage-gated sodium channel activators

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ABSTRACT

Toxins and venom components that target voltage-gated sodium (Na_v) channels have evolved numerous times due to the importance of this class of ion channels in the normal physiological function of peripheral and central neurons as well as cardiac and skeletal muscle. Na_v channel activators in particular have been isolated from the venom of spiders, wasps, snakes, scorpions, cone snails and sea anemone and are also produced by plants, bacteria and algae. These compounds have provided key insight into the molecular structure, function and pathophysiological roles of Na_v channels and are important tools due to their at times exquisite subtype-selectivity. We review the pharmacology of Na_v channel activators with particular emphasis on mammalian isoforms and discuss putative applications for these compounds.

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1. Introduction

Voltage-gated sodium (Na_v) channels are key transmembrane proteins that permit influx of Na^+ in excitable and non-excitable cells where they contribute to setting the membrane potential, action potential initiation and propagation, as well as cell motility and proliferation. While Na_v channels can be found in many tissues and cell types, they are particularly important for the function of central and peripheral neurons as well as skeletal and cardiac muscle. Nine voltage-sensing pore forming α subunit isoforms have been described in humans ($\text{Na}_v1.1$, $\text{Na}_v1.2$, $\text{Na}_v1.3$, $\text{Na}_v1.4$, $\text{Na}_v1.5$, $\text{Na}_v1.6$, $\text{Na}_v1.7$, $\text{Na}_v1.8$, and $\text{Na}_v1.9$; encoded by the genes *SCN1A*, *SCN2A*, *SCN3A*, *SCN4A*, *SCN5A*, *SCN8A*, *SCN9A*, *SCN10A* and *SCN11A*, respectively), each with distinct expression profiles. The contribution of each of these isoforms to tissue- and disease-specific physiology and pathology makes Na_v channels important therapeutic targets for treatment of a range of diseases, including epilepsy, pain, myotonias, cardiac arrhythmias and insufficiencies as well as cancer (Luiz and Wood, 2016; Mantegazza et al., 2010; Novak et al., 2015; Zaklyazminskaya and Dzemeshevich, 2016). However, most drug discovery efforts to date have been directed at the development of subtype-selective Na_v channel inhibitors.

The endogenous “activator” of Na_v channels is membrane voltage, more specifically the distribution of charge across membrane bilayers which in turn affects the position of charged transmembrane segments, leading to opening or closing of the central, Na^+ -permeable pore. Many toxins and venom components have evolved to hijack this activation mechanism by stabilising the activated conformation of the channel *via* interactions with extracellular or transmembrane domains. These compounds thus act as net channel activators, although the specific effects on the electrophysiological properties of the channel can vary significantly.

While they have been overlooked, in relative terms, for their clinical utility, Na_v channel activators not only have applications in the treatment of disease, they are also indispensable tool compounds and have provided key insights into the molecular mechanisms governing function of this important class of ion channels. Moreover, plant-derived or synthetic Na_v channel activators such as pyrethrins are one of the most commonly used classes of insecticides, with uncontrolled Na^+ influx leading to rapid spastic paralysis. In light of emerging resistance to these molecules (Silva et al., 2014), novel classes of Na_v channel activators may play important roles in the control of agricultural pests and vectors of insect-borne disease. Venom-derived peptides in particular may confer benefits as more environmentally friendly compounds that are less likely to accumulate and cause unwanted toxic effects on humans, wildlife, and beneficial insects (Li et al., 2017).

Na_v channel activators have also provided key insights into the molecular mechanisms underpinning channel activation and inactivation – key processes that are essential for normal cellular function. Future insights into the molecular and structural components involved in channel gating will undoubtedly facilitate the rational design of highly subtype-selective Na_v channel modulators with therapeutic potential.

Lastly, Na_v channel activators also represent important tool compounds that permit the dissection of the physiological and pathological roles of individual Na_v channel isoforms. They have in addition been instrumental for the development of high-throughput screening assays leading to the identification of subtype-selective Na_v channel inhibitors, as control of membrane

potential can be difficult to achieve in traditional drug discovery assays (Vetter et al., 2012a).

This review will summarise effects of various classes of Na_v channel activators, including peptide toxins from spider, wasp, snake, scorpion, cone snail and sea anemone venoms as well as alkaloids and polyethers derived from bacteria and dinoflagellates, with particular emphasis on pharmacology at mammalian Na_v channels.

2. Physiology of Na_v channels

Na_v channels are a group of sodium (Na^+) selective ion channels that are normally activated by voltage changes across cellular membranes, leading to channel opening and a subsequent transmembrane influx of Na^+ ions along their electrochemical gradient into the cell (Chandler and Meves, 1965; Hille, 1972; Hodgkin and Huxley, 1952a, b, c). Na_v channels are essential for the initiation and propagation of action potentials in electrically excitable cells like nerves and muscle fibres. They are responsible for the initial depolarization of the membrane and thus crucial for the fast electrical signalling that is necessary to propagate vital information over long distances in animals (Hodgkin and Huxley, 1952a, b, c, d).

The ion conducting α subunits can form multi-protein signalling complexes by coupling to one or two cell type specific auxiliary proteins, such as the β subunits $\beta1$ – $\beta4$ (encoded by the genes *SCN1B* – *SCN4B*), which modulate Na_v channel biophysics and trafficking (Meadows and Isom, 2005; O'Malley and Isom, 2015).

The individual Na_v channel subtypes are defined by the nine different α subunits that also contain many binding sites for toxins and other modulating compounds. Seven Na_v channels $\text{Na}_v1.1$, $\text{Na}_v1.2$, $\text{Na}_v1.3$, $\text{Na}_v1.6$, $\text{Na}_v1.7$, $\text{Na}_v1.8$, and $\text{Na}_v1.9$ play major roles in electrogenesis in neurons. The subtypes that are expressed at considerable levels in the central nervous system (CNS) include $\text{Na}_v1.1$, $\text{Na}_v1.2$, $\text{Na}_v1.3$ and $\text{Na}_v1.6$, where they can only be targeted by compounds that cross the blood-brain barrier. Widely expressed in the peripheral nervous system (PNS) are $\text{Na}_v1.1$, $\text{Na}_v1.6$, $\text{Na}_v1.7$, $\text{Na}_v1.8$, and $\text{Na}_v1.9$. Impulse electrogenesis and conduction in skeletal muscle cells and in cardiomyocytes is carried out by $\text{Na}_v1.4$ and $\text{Na}_v1.5$, respectively (Catterall et al., 2005; Goldin, 1999). Additionally, multiple Na_v channel subtypes are expressed in many non-excitable cell types like astrocytes, microglia, macrophages, and cancer cells where they play a role in e.g. the release of bioactive molecules or regulating motility, Na^+/K^+ -ATPase activity, phagocytosis and metastatic activity (Black and Waxman, 2013; de Lera Ruiz and Kraus, 2015).

Due to the important role of Na_v channels and their wide tissue distribution, it is not surprising that Na_v channel dysfunction plays an important role in numerous severe pathological conditions including epilepsy (Heron et al., 2002; Liao et al., 2010; Sugawara et al., 2001), migraine (Dichgans et al., 2005), ataxia (Liao et al., 2010; Trudeau et al., 2006), multiple sclerosis (Waxman, 2006), various painful conditions (Dib-Hajj et al., 2013; Faber et al., 2012a, 2012b; Fertleman et al., 2006; Vetter et al., 2017; Yang et al., 2004), the inability to experience pain (Ahmad et al., 2007; Cox et al., 2006), heart disorders (Medeiros-Domingo et al., 2007; Song and Shou, 2012; Tarradas et al., 2013) and neuromuscular diseases (Cannon, 2010; Jurkat-Rott et al., 2010). These conditions are often caused by inherited or spontaneous mutations, but can also be the result of injuries, adverse drug reactions or intoxications that result in hypo- and hyper-excitable phenotypes (Dib-Hajj and Waxman,

2010; Wood et al., 2004). The pharmacological modulation of any Na_V channel in the CNS, in the heart or in muscle cells has the potential risk of life-threatening side effects like seizures, cardiac arrhythmia or muscle paralysis resulting in respiratory arrest. For this reason, high potency and subtype selectivity is a much sought after characteristic in the development of pharmaceutical drugs. Both are common features of many natural toxins and venom derived peptides as a result of ongoing evolutionary specialisation and optimization.

The Na_x channel (also known as NaG or $\text{Nav}2.1$, in humans encoded by the gene *SCN7A*) also belongs to the family of the voltage-gated sodium channels, even though it is not activated by changes in membrane potential. Instead it is activated by an increased extracellular Na^+ concentration and is involved in sodium level sensing in the central nervous system, in controlling salt intake behaviour as well as in regulating epithelial sodium homeostasis (Hiyama et al., 2002; Watanabe et al., 2000; Xu et al., 2015). We use the term “voltage-gated sodium (Na_V) channel” here to mean the family of voltage activated $\text{Nav}1.\text{X}$ channels.

3. Structure of Na_V channels

3.1. Structure and function of α subunits

Structural information on Na_V channels has for a long time lagged behind functional and pharmacological characterisation. The first insights into the principal Na_V channel structure was obtained from studies with radiolabelled toxin-derivatives such as modified [^3H]saxitoxin and a [^{125}I] β -scorpion toxin that covalently bind to their receptor site. These toxins helped to identify distinct binding sites as well as the existence of one larger alpha subunit (260 kDa) and several smaller beta subunits (30–40 kDa) (Beneski and Catterall, 1980; Hartshorne et al., 1982). The same toxins also enabled the purification of a Na_V channel from the electric organ of an eel and allowed for cloning and sequence analysis of the cDNA, which revealed the amino acid sequence and thus the first primary structure of a Na_V channel (Noda et al., 1984). This started an era of educated guesswork on how the human Na_V channel structure might change during gating, and how toxins interact with the channel, that continues to date. Although we are still lacking three-dimensional high-resolution data from mammalian channels, many features can be deduced from insights from site-directed mutagenesis studies, elegant functional and pharmacological studies as well as crystal structures of the closely related bacterial channels and the cryo-EM structure of a putative Na_V channel from the American cockroach (Shen et al., 2017).

We now know that all human Na_V channel α subunits likely can form functional monomers that consist of one long polypeptide chain, which folds into four homologous but markedly different repeat domains (DI – DIV), linked by three intracellular loops, and congregate to a pseudo-tetrameric structure (Vetter et al., 2017). Major structural differences between the nine human Na_V channel α subunits $\text{Nav}1.1$ – $\text{Nav}1.9$ are unlikely, as indicated by the high sequence homology between them. In contrast, the much simpler prokaryotic Na_V channels, which have been recently used to determine Na_V channel crystal structures, are homo-tetramers, i.e. formed by four identical and separate subunits, and thus more closely resemble the structural arrangement of the mammalian K_V channels.

All Na_V channels have similar functional characteristics and a common overall structure, indicated by the high level of sequence identity with about 50–90% identical amino acids between any two isoforms (Vetter et al., 2017). However, the pharmacological and kinetic properties as well as the voltage dependence of each voltage-gated sodium channels isoform are different and

contribute to their distinct functional physiological and pharmacological roles (Catterall et al., 2005).

The characteristic features of voltage-gated sodium channels are fast gating (i.e. channel opening and closing) with voltage-dependent activation, rapid inactivation and selective sodium ion conductance (Hodgkin and Huxley, 1952d). The channels' voltage sensors move in response to the cellular membrane potential changes and thus determine the state in which the channels reside with the highest probability. Na_V channels are closed at resting membrane potentials and require membrane depolarization to be activated. Activation of the channel leads to a fast opening of the pore domain due to a conformational change, resulting in selective Na^+ ion influx into the cell and further membrane depolarization, thus contributing to the rapid upstroke of the action potential in excitable cells. The subsequent fast inactivation leads to a non-conducting channel state within milliseconds of opening (Catterall, 2012). $\text{Nav}1.9$ is an exception, taking longer (>100 ms) to inactivate than the other Na_V subtypes (Dib-Hajj et al., 2015). A special form of inactivation is the slow channel inactivation that recovers only very slowly and can occur during repetitive neuronal firing, when membranes around Na_V channels are depolarized for long periods (seconds) (Catterall, 2014).

Each of the four homologous domains DI – DIV of human Na_V channels contains six α -helical transmembrane segments (S1 – S6) and is divided into two functional domains, a pore domain (PD) and a voltage-sensing domain (VSD) (Fig. 1). The ion-conducting pore of the channel is formed by the four assembled PDs of DI – DIV, which are each shaped by segments S5 and S6 plus their extracellular linker. These four linkers each contain a conserved pore loop (P-loop) that dips halfway back into the membrane between S5 and S6, where they tightly associate with each other. This creates a narrow Na^+ selective filter formed by the amino acids DEKA and EEDD at the extracellular end of the pore. The pore region presents toxin binding sites on the external opening, close to the selectivity filter (for pore blockers like tetrodotoxin and the μ -conotoxins) as well as inside the pore (for gating modifiers like veratridine) (de Lera Ruiz and Kraus, 2015).

The four functional and structurally distinct VSDs are each shaped by the segments S1–S4 (Catterall, 2000; Guy and Seetharamulu, 1986) and linked to the PD via the S4–S5 linker. The S4 segments function as voltage sensors of the channel as they carry several highly conserved positively charged arginine and lysine residues (gating charges) that are forced to move across the cell membrane upon changes in the membrane potential and induce subsequent changes in the channel conformation (Guy and Seetharamulu, 1986; Stuhmer et al., 1989). Especially the two small linkers between S1–S2 and S3–S4, but also residues in the transmembrane segments of the VSDs, present further extracellular binding sites for many gating modifier toxins like spider and scorpion toxins (de Lera Ruiz and Kraus, 2015; Murray et al., 2016).

Another important structural feature of the Na_V channel α subunits is the third intracellular loop (L3) containing an IFM motif that connects S6 in DIII with S1 in DIV and is crucial for inactivation of the channel (Fig. 1).

While there is evidence for the importance of DIV for fast inactivation, the distinct functional roles of the three other domains is still uncertain (Ahern et al., 2016), partly due to a probable functional coupling between the different domains and a lack of more detailed structural information.

Recent advances in single particle cryo-electron microscopy and membrane protein crystallography have improved our general structural understanding of voltage-gated ion channels, including the mammalian calcium and potassium channel (Hite et al., 2015; Whicher and MacKinnon, 2016; Wu et al., 2015) and bacterial as well as eukaryotic Na_V channels (Catterall, 2014; Shen et al., 2017).

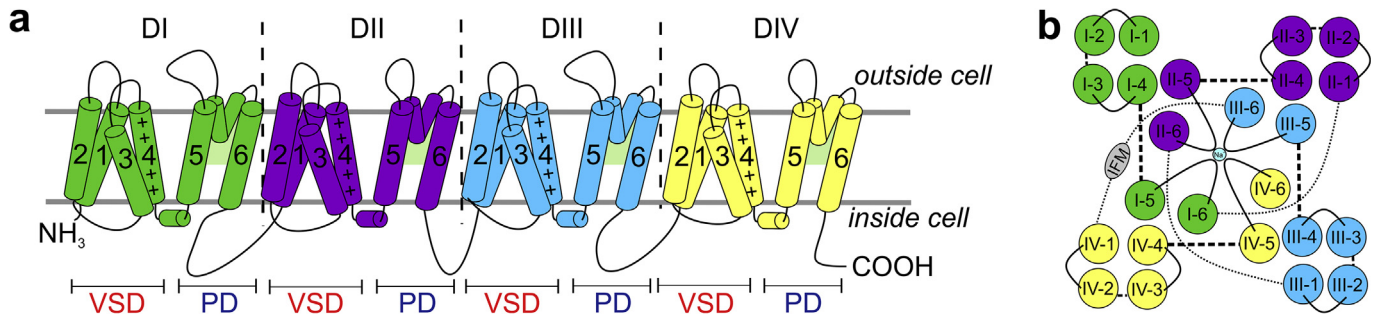


Fig. 1. General structural topology of Na_v channels. **a)** Na_v channels consist of four homologous domains (domain I, II, III and IV) that are linked via large intracellular loops. Each domain contains 6 transmembrane segments (S1 – S6) which can be functionally divided into the voltage-sensing domains (VSD; segments S1–S4) and the pore domains (PD, segments S5–S6). The S4 segments carry several highly conserved positively charged arginine and lysine residues (gating charges) which move across the cell membrane upon changes in the membrane potential and thus function as the voltage sensors of the channel. **b)** Top view of the Na_v channel. The ion-conducting pore of the channel is formed by the four assembled PDs of DI – DIV, which are each shaped by segments S5 and S6 plus their extracellular linker. The arrangement of the voltage-sensing (S1–S4) and pore-forming (S5–S6) segments are offset so that in functional channels, the voltage sensor of each domain is closest to the pore-forming segment of the following domain. Broken lines indicate intracellular linkers and loops; continuous lines indicate extracellular linkers.

However, the exact three-dimensional structure of the pseudo-tetrameric human voltage-gated sodium channel α subunits is still unknown due to difficulties with its crystallization. Several more or less accurate homology models exist, which are based on human Na_v channel sequences and the X-ray crystal structures of homo-tetrameric prokaryotic Na_v channels (NavAb, NavRh, and NavMs) (Catterall, 2014; McCusker et al., 2012; Payandeh et al., 2011, 2012; Zhang et al., 2012). One of the problems with these models is that the much simpler bacterial channels lack several important features of the eukaryotic version. Specifically, the four assembled identical subunits do not have the intracellular loops that connect the four corresponding human domains. Accordingly, the third loop (L3) with the conserved amino acid residues IFM between DIII and DIV, which is essential for fast inactivation of the open mammalian Na_v channel (Stuhmer et al., 1989; Vassilev et al., 1989; West et al., 1992), is not conserved. Furthermore, a symmetrical bacterial channel cannot explain the distinct roles of the four individual domains, which each have different kinetics, functional properties and contributions to channel gating as well as unique binding sites for interacting compounds (Ahern et al., 2016). Another general problem of crystallography studies is the rigidity of the crystallized channels, which makes it difficult to analyse the *in vivo* flexible extracellular loops that determine the binding site of many channel modulators. Nevertheless, information on the general design of the Na_v channels obtained from these studies is still helpful for a better understanding of these important proteins. With the continuing high interest in the field of Na_v channels, it is likely that rapid advances in the field of high resolution single particle cryo-electron microscopy will lead to additional structural details on mammalian Na_v channels in the near future (Sato et al., 2001; Slowik and Henderson, 2015).

3.2. Structure and function of β subunits

When α subunits are heterologously expressed alone, they demonstrate full Na_v channel function including sodium selectivity, rapid opening and fast inactivation (Goldin et al., 1986; Noda et al., 1986). However, *in vivo* they typically couple to one or two cell-type specific Na_v channel β subunits, which do not contribute to the ion-conducting pore but modulate channel function. This makes the coexpression of the appropriate β subunits in heterologous expression systems necessary to mimic the kinetic properties of native sodium channels. β Subunits modify Na_v channel gating properties including kinetics and voltage dependence as well as channel expression and trafficking (Catterall, 2000; Meadows and Isom, 2005; O'Malley and Isom, 2015). However, β subunits do

not only modify sodium currents but have additional functions as cell adhesion molecules, playing roles in cell adhesion, migration, invasion and neurite outgrowth (Brackenbury and Isom, 2011; Patino and Isom, 2010). Furthermore, the β_4 subunit has been found to function as an endogenous cytoplasmatic open channel blocker that directly competes with the fast inactivation gate and thus enabling resurgent currents, which render certain Na_v channel isoforms capable of high-frequency firing in excitable tissues (Lewis and Raman, 2013).

The identified genes SCN1B, SCN2B, SCN3B and SCN4B encode the four membrane bound β subunits β_1 – β_4 as well as a fifth soluble splice variant β_1b . All are glycoproteins with a molecular weight of about 30–40 kDa, consisting of one single transmembrane domain (which is lacking in β_1b) and an immunoglobulin-like extracellular domain. The subunits β_1 and β_3 interact non-covalently with α subunits, whereas β_2 and β_4 bind via disulfide bonds covalently to Na_v channel α subunits.

β Subunits are involved in several Na_v channel-dependent pathologies, with changed expression levels in conditions like pain and nerve injury and genetic mutations in all four β subunit-encoding genes being linked to severe diseases (Eijkelkamp et al., 2012).

Interestingly, changing expression levels of β subunits in heterologous expression systems can significantly affect affinity and efficacy of activating and inhibitory toxins and thus alter Na_v channel modulation (Gilchrist et al., 2013; Namadurai et al., 2015). For instance, the presence of the β_4 subunit dramatically reduces the inhibition of Na_v1.2 by the spider toxin ProTx-II and decreases the maximal conductance of the same channel induced by the activating β -scorpion toxin TsVII, while not affecting its hyperpolarizing shift in voltage of activation. Co-expression of the Na_v channel β_2 and β_1 subunit significantly reduces the activation of Na_v1.2 channels by the sea anemone toxin ATX-II and the α -scorpion toxin LqIV, respectively (Gilchrist et al., 2013). β Subunit co-expression can also alter the kinetics of Na_v channel inhibition by pore blocking modulators of the conotoxin family. Binding on-rates (k_{on}) of μ -conotoxins at several Na_v channel isoforms can be increased in the presence of β_1 and β_3 subunits, while they are decreased by β_2 and β_4 subunits (Zhang et al., 2013). In contrast, all four β subunits increase the k_{on} of μ O-conotoxin MrVIB at Na_v1.8 channels (Wilson et al., 2011). Furthermore, co-expression of β_2 or β_4 protects Na_v1.1–1.7 (excluding Na_v1.5) channels against block by an analogue of the μ O β -conotoxin GVIIJ (Gajewiak et al., 2014). Thus, coexpression of β subunits can strongly influence the affinity of conopeptides for Na_v channels.

Modification of toxin pharmacology by the presence of β

subunits may thus explain some tissue-specific differences in pharmacological activity, as the expression levels of β subunits vary in different cell types (Gilchrist et al., 2013; Wilson et al., 2011). In addition, the expression of β subunits has implications for the use of Na_v channel modulators in pathological conditions where β subunit expression is changed.

3.3. Gating mechanisms of Na_v channels: a toxin perspective

Na_v channels exist in at least three, and likely many more, distinct states: resting (closed), open, and inactivated (Fig. 2). The process of channel gating refers to the movement of the voltage sensors in response to changes in charge distribution across the cellular membrane, resulting in structural re-arrangement of the channel. Toxins have been indispensable tools to understand the complex transitions between these states, and generally speaking interact with Na_v channels either by physically occluding the pore, or by preventing movement of the voltage sensors either in the outward or inward direction. Accordingly, to understand the pharmacological effects and mechanism of action of toxins, it is imperative to understand the gating mechanisms of Na_v channels.

In the resting or closed state of Na_v channels, all four VSD are presumably in the “down” state as their positive gating charges on the S4 segments are forced toward the intracellular side of the cell membrane. This is due to the strong electrical potential generated by the negative resting membrane potential of unexcited neurons and myocytes. Membrane depolarization leads to activation and very quick opening of the channel as the more positive membrane potentials allow the S4 gating charges to move toward the

extracellular surface into the “up” position. This likely involves a “helical screw” motion in which S4 helices slide and rotate around their axes, with the charged residues exchanging ion-pair partners until they are stabilised again by highly conserved negative charge clusters on the other segments (Chanda and Bezanilla, 2002; Guy and Seetharamulu, 1986; Payandeh et al., 2011). The movement in VSDs is subsequently transferred to the PDs via the S4 - S5 linkers, changing channel conformation and allowing the pore to become permeable to sodium ions (Li et al., 2014; Payandeh et al., 2011).

Fast inactivation follows channel opening, resulting in a non-conducting inactivated channel state before finally recovering from inactivation and returning to a resting, closed state. Crucial for fast inactivation are the highly conserved residues IFM (isoleucine-phenylalanine-methionine) on the third intracellular loop between DIII and DIV (Vassilev et al., 1989; West et al., 1992), which bind to an unknown receptor site involving several residues of the S4 - S5 linker of DIV, close to the intracellular opening of the pore and thus closing it. The highly conserved DIV S4 - S5 linker is made up of 15 amino acids of which 14 are identical in all nine human Na_v channel isoforms. At least 7 of these residues have been found to be important for normal fast inactivation in various mutagenesis studies and have thus been proposed to be part of the putative receptor site of the docking region with the IFM motif (Filatov et al., 1998; McPhee et al., 1998; Mitrovic et al., 1996; Tang et al., 1996).

It is possible that there is functional coupling between the four VSDs that affects channel gating, but this is still unclear. Mutagenesis studies on S4 gating charges in the four VSDs hinted at a predominant role of DIV in fast inactivation (Kuhn and Greeff, 1999), while activation gating was affected by mutations of

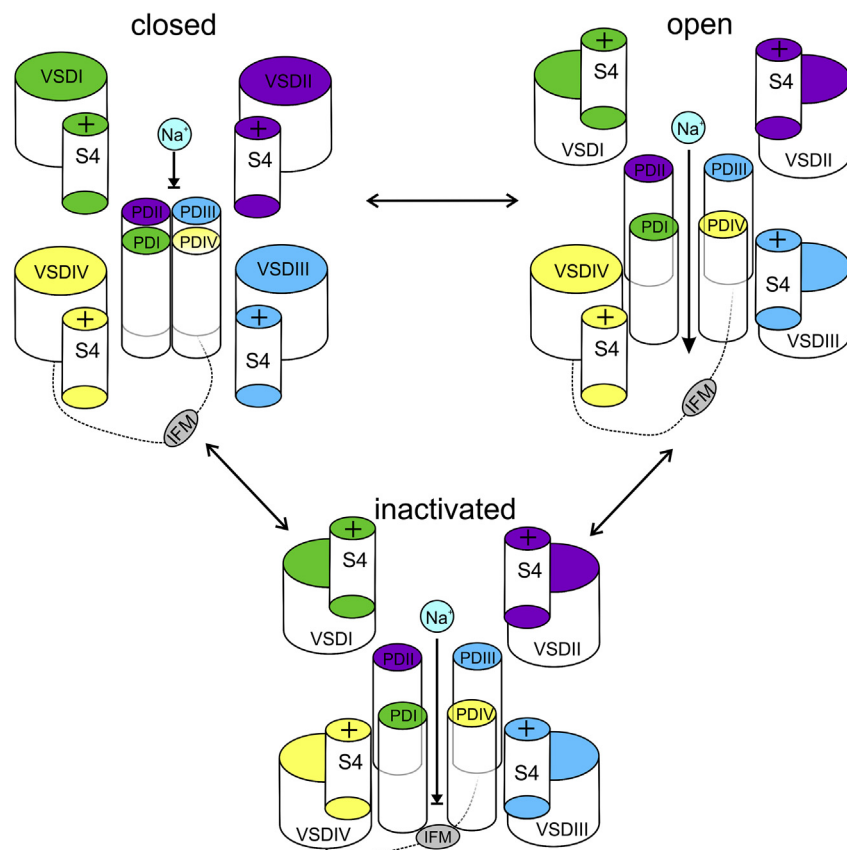


Fig. 2. Simplified gating model of Na_v channels. Na_v channels exist in at least three, and likely many more, distinct states: resting (closed), open, and inactivated. In the resting or closed state of Na_v channels, all four VSD are presumably in the inactivated state. Movement of the DI-DIII VSDs, which is subsequently transferred to the PDs via the S4 - S5 linkers, changes channel conformation and allows the pore to become permeable to sodium ions. Channel inactivation requires movement of the domain IV VSD, which brings the IFM motif on the third intracellular loop between DIII and DIV close to the intracellular opening of the pore and thus inactivates it.

positive S4 charges in all four voltage-sensing domains, but predominantly DI–DIII (Kontis et al., 1997). Structure-function studies with toxins that bind to the DIV S3–S4 linker of the VSD confirmed the role of DIV in fast inactivation (Benzinger et al., 1998; Rogers et al., 1996). The results suggest that fast inactivation can only occur after a conformational change in the VSD of domain IV makes it move toward the extracellular side of the cell membrane, while fast inactivation is independent from the movement of the VSDs of DI–DIII, which are likely to be mainly responsible for initial channel opening. Consistent with these findings is the observation that upon channel activation, the VSDs in DI–DIII move faster toward the extracellular surface with time-constants correlating to the fast channel opening, while the movement of the VSD in DIV is slower and correlates with the subsequent fast inactivation (Chanda and Bezanilla, 2002). Moreover, Na_v channel modulating toxins that only bind to the DIV VSD usually have no effects on channel activation but inhibit or delay fast inactivation, resulting in slower decaying sodium currents. However, the separate functions of the kinetically different voltage sensors in DI–DIII still remain to be defined. The DII and DIV VSDs are targeted by many natural modulators, but only very few toxins affect DI and DIII with binding sites rarely reported for these two domains (Bosmans et al., 2008). This suggests a predominant role of DII in Na_v channel activation as it was probably evolutionarily the most successful to target.

4. Na_v activator pharmacology

Due to its important role in action potential generation and propagation, many clinical drugs have been developed and numerous toxins from venomous animals or plants have evolved to target Na_v channels, often as part of prey capture or defence strategies (Kaczorowski et al., 2008; Kalia et al., 2015). At least eight distinct binding sites on Na_v channels (site 1–8) have been proposed, six of which are the receptor sites for various toxins (Catterall et al., 2005; de Lera Ruiz and Kraus, 2015). The definition of these binding sites is often complex as they include crucial amino acids on multiple domains and segments of the channel that

significantly overlap with other binding sites. In a simplified model, binding sites can be differentiated structurally into the pore region and the voltage-sensing domains, with additional elaboration on functional effects distinguishing compounds that enhance or decrease channel activation, as well as those that enhance or decrease channel inactivation. According to the resultant effects on macroscopic Na_v channel currents, these compounds can thus be classified as channel inhibitors or activators (Fig. 3).

Specifically, the pore region binding modulators can be functionally divided into pore occluding pore blocker (like tetrodotoxin and lidocaine) and compounds that facilitate or prolong channel activation by modifying gating after binding into the pore (e.g. veratridine). The VSD binding modulators can be broadly divided into another two groups. The first group – functional Na_v channel activators – leads to an increase in macroscopic Na^+ current which can be induced by facilitating activation (hyperpolarizing shift in voltage of activation) and/or prolonging the open channel state (inhibition of inactivation). Examples of this group are discussed in more detail in the following sections. The second group – functional Na_v channel inhibitors – leads to a decrease in Na^+ current, induced by reduced activation (depolarising shift in voltage of activation; e.g. ProTx-II) and/or shortened open channel states after binding to a VSD (e.g. Pn3a) (Deuis et al., 2017).

Na_v channel activators have provided crucial tool compounds to delineate the physiological and pathological roles of Na_v channels, and may in future find applications as therapeutics for treatment of conditions such as epilepsy and cardiac arrhythmias. However, in contrast to Na_v channel inhibitors, detailed pharmacological characterisation of Na_v channel activators is relatively sparse, in part because more complex electrophysiological protocols may be required to determine effects on diverse gating parameters. Accordingly, it can be difficult to accurately define EC_{50} values for Na_v channel activators, which depend on distinct, and finite, effects on a single parameter.

Na_v channel activator toxins are typically gating modifiers toxins that bind either to the inside of the pore or to the extracellular parts of DII VSD and/or DIV VSD, sometimes involving

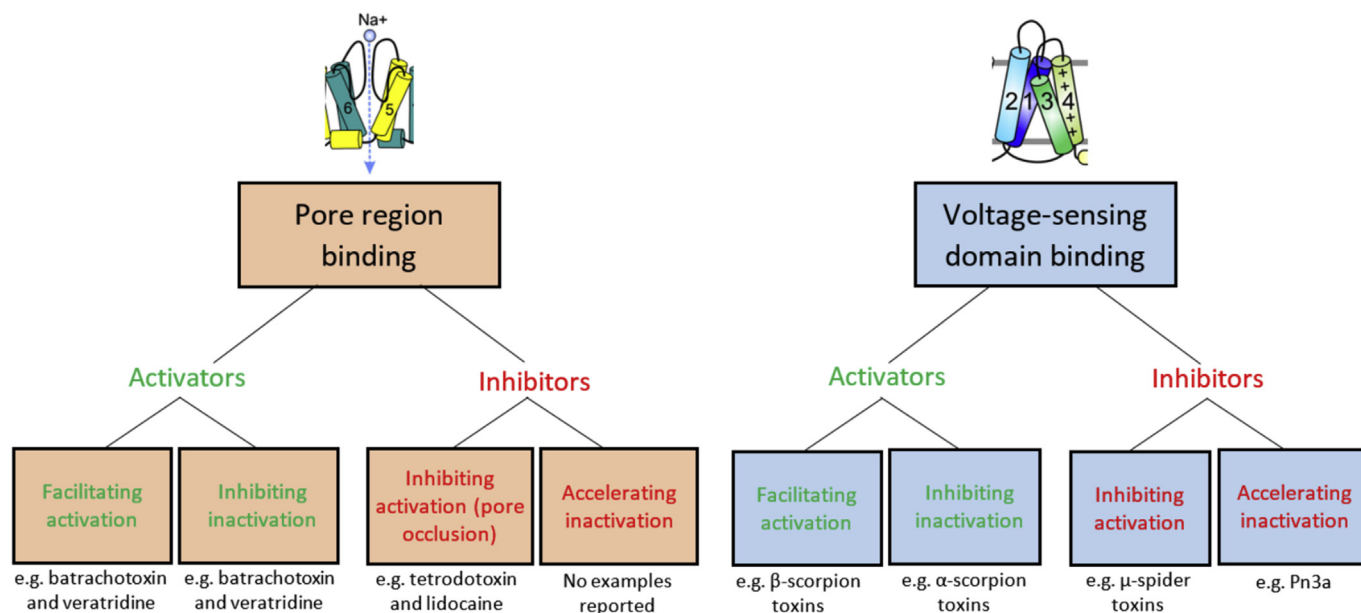


Fig. 3. Simplified classification of Na_v channel modulators. In a simple approach, the binding sites of Na_v channel modulators can be divided into the pore region (consisting of all four S5 and S6 segments including their linkers) and the four voltage-sensing domains (consisting of segments S1 to S4 including their extracellular linkers). Further functional classification of activators and inhibitors results in four possible classes of Na_v channel modulators for each binding site. Some Na_v channel modulators belong to more than one class and can have differential functional effects on distinct channel isoforms. Given that binding to the VSDs of different domains likely preferentially affects distinct channel functions, the number of possible classes of Na_v modulators may be even greater.

extracellular pore domain residues in close proximity to the VSDs (Murray et al., 2016; Osteen et al., 2016). Gating modifiers modulate conformational changes of Na_v channels and the coupling between the different channel states with varying effects on activating and inactivating gating properties. In general terms, activator toxins binding to DII VSD generally affect activation by shifting the voltage of activation to more hyperpolarized potentials, while activator toxins that bind solely to the VSD of DIV generally affect inactivation by inhibiting normal movement of S4 and the subsequent inactivation of the channel (Xiao et al., 2014).

4.1. Classes of activators

Due to their evolutionary success – based on the serious physiological effects caused by uncontrolled Na_v channel activity – Na_v channel activators are found in many different organisms, including plants, algae, frogs, cone snails, spiders, wasps, snakes, sea anemone and scorpions. In addition to alkaloid and polyether Na_v channel activators, many of the venom-derived Na_v channel activators are peptidic in nature, although the size, sequence and structure of these compounds is diverse. The Na_v channel selectivity and toxin-channel interactions for each class of activators are discussed below, as well as being summarised in Tables 1 and 2, respectively. It should be noted that potency might differ between human and rodent orthologs of Na_v channels, however this is

generally not assessed systematically.

4.1.1. Alkaloid Na_v channel activators

Batrachotoxin, grayanotoxin, veratridine and aconitine are a chemically diverse group of Na_v channel activators that share an overlapping binding site, designated as site 2 (Catterall, 1975). While often grouped together, these alkaloids have differential effects on the biophysical properties of Na_v channels, all of which will be discussed individually below.

Batrachotoxin, first isolated in 1965, is a lipophilic alkaloid secreted by the skin glands of poison dart frogs (*Phyllobates* spp.), used by native South Americans to poison the tips of blow-darts (Daly et al., 1965). Batrachotoxin is not produced by the frog itself, but is bioaccumulated through dietary consumption of alkaloids from insects, although the exact dietary source remains unknown (Daly et al., 1994). Accordingly, frogs raised in captivity do not secrete batrachotoxin, and due to difficulties in chemical synthesis, which were only recently overcome (Logan et al., 2016), thousands of the now endangered *Phyllobates* poison dart frogs were captured and killed to isolate batrachotoxin (Daly et al., 1994). Symptoms of batrachotoxin intoxication include cardiac arrhythmias, motor impairment, respiratory depression and convulsions, which can lead to death (Albuquerque et al., 1971).

Batrachotoxin preferentially binds to the open state of Na_v channels, requiring repetitive depolarising pulses (>1000) at a

Table 1

Potency and selectivity of Na_v activator toxins at Na_v1.1–1.9. Where available the EC₅₀ values are provided. If not available activity is classified as either sensitive (S) or insensitive (IS) at the concentration stated.

Toxin	Na _v 1.1	Na _v 1.2	Na _v 1.3	Na _v 1.4	Na _v 1.5	Na _v 1.6	Na _v 1.7	Na _v 1.8	Na _v 1.9	Ref
Alkaloid										
Batrachotoxin	NA	S (10 μM)	S (10 μM)	S (10 μM)	S (10 μM)	S (10 μM)	S (10 μM)	S (10 μM)	NA	(Bosmans et al., 2004; Li et al., 2002; Logan et al., 2016)
Spider										
δ-hexatoxin-Mg1a	S (5 μM)	S (5 μM)	S (5 μM)	IS (5 μM)	IS (5 μM)	S (5 μM)	S (5 μM)	IS (5 μM)		(Yamaji et al., 2009)
δ-theraphotoxin-Cg3a	NA	NA	2 μM	>5 μM	125 nM	NA	>5 μM	NA	NA	(Huang et al., 2015)
δ-theraphotoxin-Cg1a	NA	870 nM	845 nM	339 nM	335 nM	NA	348 nM	NA	NA	(Tao et al., 2016)
δ-theraphotoxin-Hm1a	38 nM	236 nM	220 nM	IS (100 nM)	IS (100 nM)	IS (100 nM)	IS (100 nM)	IS (100 nM)	IS (100 nM)	(Osteen et al., 2016)
Wasp										
β-pompilidotoxin	S (46 μM)	21 μM	S (46 μM)	IS (140 nM)	IS (140 nM)	30 μM	S (46 μM)	NA	NA	(Schiavon et al., 2010)
α-Scorpion										
OD1	NA	5 μM	3 μM	10 nM	4 μM	47 nM	7 nM	IS (1 μM)	NA	(Durek et al., 2013b; Maertens et al., 2006a)
Ts2	NA	S (1 μM)	S (1 μM)	IS (1 μM)	S (1 μM)	S (1 μM)	S (1 μM)	IS (1 μM)	NA	(Cologna et al., 2012)
MeuNaTxα-1	NA	S (1 μM)	S (1 μM)	IS (1 μM)	IS (1 μM)	3 μM	S (1 μM)	IS (1 μM)	NA	(Zhu et al., 2012)
MeuNaTxα-2	NA	IS (1 μM)	IS (1 μM)	2 μM	IS (1 μM)	IS (1 μM)	IS (1 μM)	IS (1 μM)	NA	(Zhu et al., 2012)
MeuNaTxα-5	NA	IS (2 μM)	S (2 μM)	S (2 μM)	S (2 μM)	0.79 μM	S (2 μM)	IS (2 μM)	NA	(Zhu et al., 2012)
MeuNaTxα-12	0.91 μM	S (10 μM)	NA	S (10 μM)	IS (10 μM)	S (10 μM)	NA	NA	NA	(Zhu et al., 2013)
MeuNaTxα-13	2.5 μM	S (10 μM)	NA	S (10 μM)	IS (10 μM)	S (10 μM)	NA	NA	NA	(Zhu et al., 2013)
β-Scorpion										
Cn2	IS (300 nM)	IS (300 nM)	IS (300 nM)	IS (300 nM)	IS (300 nM)	40 nM	IS (300 nM)	NA	NA	(Schiavon et al., 2006)
Tz1	NA	S (10 μM)	NA	IS (10 μM)	IS (10 μM)	S (10 μM)	IS (10 μM)	NA	NA	(Leipold et al., 2006)
Tf2	IS (1 μM)	IS (1 μM)	S (1 μM)	IS (1 μM)	IS (1 μM)	IS (1 μM)	IS (1 μM)	IS (1 μM)	NA	(Camargos et al., 2015)
Ts1	IS (100 nM)	IS (100 nM)	S	S	S	S	IS	IS	NA	(Peigneur et al., 2015b)
Cone snail										
δ-SuVIA	NA	NA	4 nM	5 nM	1 nM	2.5 nM	NA	NA	NA	(Jin et al., 2015)
ι-RXIA	IS (50 μM)	18 μM	IS (50 μM)	IS (50 μM)	IS (50 μM)	1.8 μM	S (50 μM)	IS (50 μM)	IS (50 μM)	(Fiedler et al., 2008)
Sea anemone										
ATX-II	6 nM	8 nM	759 nM	109 nM	49 nM	~180 nM	1.8 μM	NA	NA	(Oliveira et al., 2004; Wanke et al., 2009)
AFT-II	391 nM	2 μM	459 nM	31 nM	63 nM	~300 nM	5.8 μM	NA	NA	(Oliveira et al., 2004; Wanke et al., 2009)
Bc-III	~300 nM	1.5 μM	1.5 μM	821 nM	307 nM	~900 nM	5.7 μM	NA	NA	(Oliveira et al., 2004; Wanke et al., 2009)
CGTX-II	165 nM	>1 μM	>1 μM	>1 μM	105 nM	133 nM	IS (50 nM)	NA	NA	(Zaharenko et al., 2012)
Dinoflagellate										
P-CTX-1	8 nM	8 nM	3 nM	18 nM	10 nM	18 nM	13 nM	2 nM	S (1 nM)	(Inserra et al., 2017)

NA; not assessed.

Table 2

Summary of experimentally determined Nav channel interactions with Nav activator toxins. For simplicity, only the S1–S2 and S3–S4 linkers or the S5 and S6 transmembrane segments (or S5–S6 linker as indicated) are presented for DI, DII, DIII and DIV. The type of experimental evidenced is abbreviated as follows: single-point channel mutations (S); chimeric channels (C); competitive radioligand binding (R); photoaffinity labelling (P). Letters alone indicate an experimentally determined interaction, ? indicates that it is unknown whether there is an interaction, and - indicates that it has been experimentally determined that there is no interaction.

	DI				DII				DIII				DIV				Ref	
	S1	S3	S5	S6	S1	S3	S5	S6	S1	S3	S5	S6	S1	S3	S5	S6		
	–S2	–S4			–S2	–S4			–S2	–S4			–S2	–S4				
Alkaloids																		
Batrachotoxin	?	?	?	S	?	?	?	S	?	?	?	S	?	?	?	S	(Toma et al., 2016; Wang et al., 2001; Wang et al., 2000b; Wang and Wang, 1998, 1999)	
Grayanotoxin	?	?	?	S	?	?	?	?	?	?	?	?	?	?	?	S	(Ishii et al., 1999; Kimura et al., 2000)	
Veratridine	?	?	?	S	?	?	?	?	?	?	?	?	?	?	?	S	(Wang et al., 2000a)	
Spider																		
δ-hexatoxin-Ar1a	?	?	R ^{a,d}	?	?	?	?	?	?	?	?	R ^a	R ^a	?	?		(Little et al., 1998)	
δ-hexatoxin-Hv1a	?	?	R ^{a,d}	?	?	?	?	?	?	?	?	R ^a	R ^a	?	?		(Little et al., 1998)	
β-hexatoxin-Mg1a			R ^{b,d}	R ^b	R ^b							R ^{b,d}	R ^a				(Corzo et al., 2003)	
δ-theraphotoxin-Cg1a	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Tao et al., 2016)	
δ-theraphotoxin-Cg3a	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Huang et al., 2015)	
δ-theraphotoxin-Hm1a	?	–C	?	?	?	–C	?	?	?	–C	?	?	–C	–C	?	?	(Osteen et al., 2016)	
Wasp																		
β-pompilidotoxin	–C	–C	–C ^d	–C	–C	–C ^d	–C	–C	–C	–C	–C	S	–C ^d				(Kinoshita et al., 2001)	
α-Scorpion																		
LqTx	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Rogers et al., 1996)	
LqhII	?	?	S ^d	?	?	?	?	?	?	?	?	S	?	?	?		(Leipold et al., 2005; Leipold et al., 2004; Wang et al., 2011)	
LqhIII	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Leipold et al., 2005; Leipold et al., 2004)	
β-Scorpion																		
CssIV	–C		–C ^d	S	S	–C ^d	–C	–C	S ^d	–C	–C	–C ^d					(Cestele et al., 1998; Cestele et al., 2006; Zhang et al., 2011)	
Tz1	–C	–C	–C ^d		S				S ^d	–C	–C	–C ^d					(Leipold et al., 2006)	
Cone snail																		
δ-SVIE	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Leipold et al., 2005)	
Sea anemone																		
ATX-II	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Rogers et al., 1996)	
ApB	?	?	?	?	?	?	?	?	?	?	?	?	S	?	?		(Benzinger et al., 1998)	
Dinoflagellate																		
Brevetoxin	?	?	?	P	?	?	?	?	?	?	?	?	?	P ^d			(Trainer et al., 1994)	
Ciguatoxin	?	?	?	R ^c	?	?	?	?	?	?	?	?	?	R ^{c,d}			(Lombet et al., 1987)	

^a Competitive radioligand binding with LqhII.

^b Competitive radioligand binding with CssiV.

^c Competitive radioligand binding with brevetoxin.

^d S5–S6 linker.

frequency of 1–2 Hz to observe a functional effect under voltage-clamp conditions (Wang and Wang, 1998). Once bound, batrachotoxin causes a hyperpolarizing shift in the voltage-dependence of activation, inhibits fast and slow inactivation, and causes a change in ion selectivity (Bosmans et al., 2004; Li et al., 2002; Wang and Wang, 1996, 1998). Batrachotoxin has an EC₅₀ of 1 μM, and displays little Nav channel selectivity, inhibiting fast inactivation of Nav1.2, Nav1.3, Nav1.4, Nav1.5, Nav1.7 and Nav1.8 with a similar effect at a concentration of 10 μM (Bosmans et al., 2004; Li et al., 2002; Logan et al., 2016).

The binding site of batrachotoxin is located within the inner pore region, with single point mutations of amino acids located on S6 in DI (I433, N434, L437), DII (N784, L788), DIII (F1236, S1276, L1280) and DIV (F1579, N1584) causing rNav1.4 to become insensitive to the effects of batrachotoxin (Figs. 4–8) (Toma et al., 2016; Wang et al., 2000b, 2001; Wang and Wang, 1998, 1999). These amino acids are conserved on hNav1.1–1.8, accounting for the limited selectivity of batrachotoxin for Nav channels. Interestingly, sequence alignment of frog Nav1.4 has identified similar single point mutations in *Phylllobates* poison dart frogs on S6 DI (S429A, I433V, A446D) and S6 DIV (V1583I, N1584T), conferring resistance to protect against self-intoxication (Tarvin et al., 2016).

Grayanotoxin is present in the leaves, twigs and flowers of *Rhododendron* plants. In humans, intoxication can occur by consumption of honey (known as “mad honey”) produced by bees from the nectar of *Rhododendron* plants, with most reported cases

occurring from honey originating from Turkey (Gunduz et al., 2006). Mad honey poisoning causes nausea, vomiting, dizziness, hypotension, bradycardia and other cardiac disturbances, which can be fatal if left untreated.

Like batrachotoxin, grayanotoxin binds to the open state of Nav channels and requires repetitive depolarising pulses to see a functional effect under voltage-clamp conditions (Yakehiro et al., 2000). Once bound, grayanotoxin causes a hyperpolarizing shift in the voltage-dependence of activation, but unlike batrachotoxin, causes a reduction in peak current (Kimura et al., 2001). Grayanotoxin also inhibits fast inactivation and causes a reduction in ion selectivity, similar to batrachotoxin (Yuki et al., 2001). Compared with batrachotoxin, grayanotoxin is less potent, with an EC₅₀ of 31 μM, and has a greater effect on fast inactivation at Nav1.4 compared to Nav1.5, although activity at other Nav channel subtypes in heterologous expression systems remains to be determined (Kimura et al., 2001; Yakehiro et al., 2000).

The binding site of grayanotoxin is less well defined, with studies focused on assessing the activity of grayanotoxin on rNav1.4 channels with single point mutations known to cause batrachotoxin insensitivity. While single point mutations of amino acids located on S6 in DI (I433, N434, L437) cause grayanotoxin insensitivity, activity is maintained with the single point mutation F1579A on S6 in DIV, indicating that grayanotoxin and batrachotoxin share an overlapping but non-identical binding site (Fig. 4)

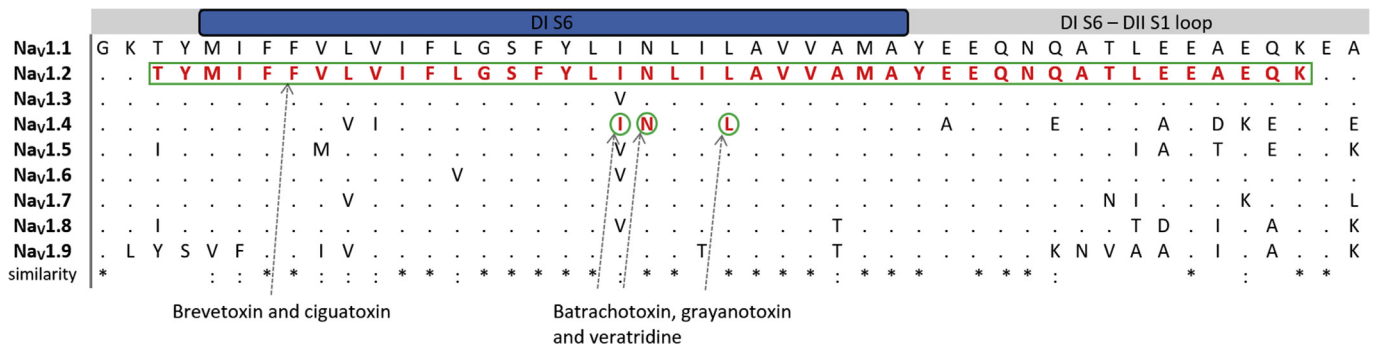


Fig. 4. Amino acid sequence alignment of human Nav isoforms for DI S6. Multiple amino acid sequence alignment of domain I segment 6 and adjacent residues of hNav α subunit isoforms was performed using Clustal Omega 1.2.2 with default parameters. We used the amino acid sequences in the canonical isoform 1 as provided by the UniProtKB database (www.uniprot.org/). UniProtKB entry identifiers: P35498 (SCN1A, hNav_v1.1); Q99250 (SCN2A, hNav_v1.2); Q9NY46 (SCN3A, hNav_v1.3); P35499 (SCN4A, hNav_v1.4); Q14524 (SCN5A, hNav_v1.5); Q9UQD0 (SCN8A, hNav_v1.6); Q15858 (SCN9A, hNav_v1.7); Q9Y5Y9 (SCN10A, hNav_v1.8); Q9UI33 (SCN11A, hNav_v1.9). The definition of transmembrane segments and intra- and extracellular domains are as specified by the UniProtKB database. Where dots are shown instead of letters in Nav isoform sequences, the amino acid is identical to the one in Nav_v1.1. Consensus symbols: * = positions which have a single, fully conserved residue in all sequences; = = conservation between groups of strongly similar properties in all sequences (STA; NEQK; NHQK; NDEQ; QHRK; MILV; MILF; HY; FYW); = = conservation between groups of weakly similar properties in all sequences (CSA; ATV; SAG; STNK; STPA; SGND; SNDEQK; NDEQHK; NEQHRK; FVLIM; HFY). Amino acid residues important for Nav channel activator toxin binding are in bold. Residues are additionally circled if evidence is based on channel mutagenesis studies and sequence parts are highlighted in rectangles if evidence is based on studies with chimeric channels. As several studies have been carried out on rat Nav channels, the results are only indicated if the human isoform has the same residues at the corresponding positions. Brevetoxin, ciguatoxin, batrachotoxin, grayanotoxin and veratridine have been shown to bind to residues in the DI S6 region.

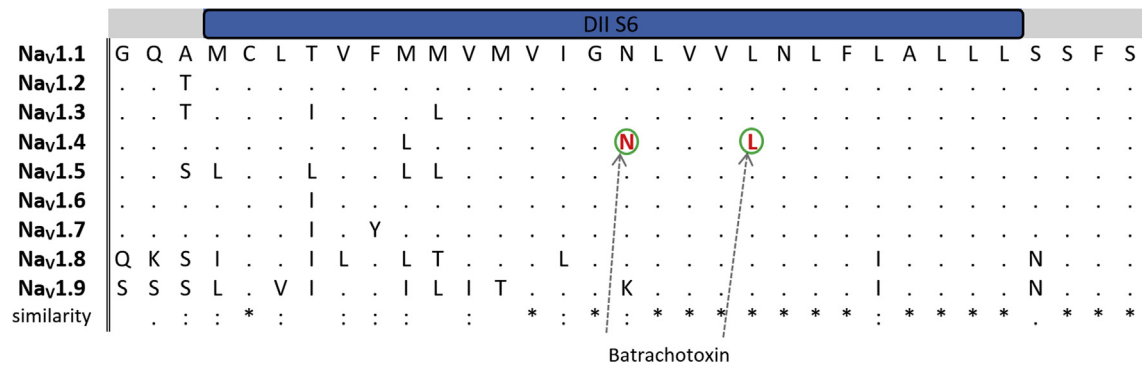


Fig. 5. Amino acid sequence alignment of human Nav isoforms for DII S6. Multiple sequence alignment of domain II segment 6 and adjacent residues was performed as stated in Fig. 4. Batrachotoxin binds to residues in DII S6.

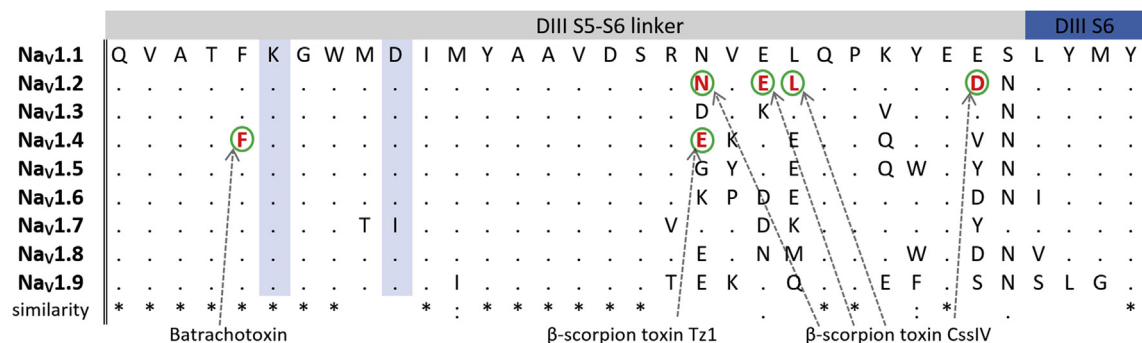


Fig. 6. Amino acid sequence alignment of human Nav isoforms for DIII S5 - S6 linker. Multiple sequence alignment of domain II S5 - S6 linker and adjacent residues was performed as stated in Fig. 4. The shaded conserved lysine and aspartic acid residues are part of the channels sodium selectivity filter in the P-loop of DIII. This linker contains residues important for interactions with batrachotoxin as well as the β-scorpion toxins CsslV and Tz1.

(Ishii et al., 1999; Kimura et al., 2000). In addition, the poison dart species *Phyllobates terribilis* is insensitive to batrachotoxin, but not to grayanotoxin, confirming a non-identical binding site (Daly et al., 1980).

Veratridine is a steroid-derived alkaloid found in the roots of *Veratrum* plants from the *Liliaceae* family. Unlike batrachotoxin and

grayanotoxin, veratridine does not require repetitive depolarising pulses to exert a functional effect on Nav channels under voltage-clamp conditions (Farrag et al., 2008; Ghatpande and Sikdar, 1999). Veratridine causes a hyperpolarising shift in the voltage-dependence of activation and decreases peak current, however compared to batrachotoxin and grayanotoxin, only delays rather

than inhibits fast inactivation, resulting in a comparatively small persistent current (Ghatpande and Sikdar, 1999). Veratridine has an EC₅₀ of 29 μM at Nav1.7, and appears to have minimal selectivity at Nav1.1–1.7, based on its use as a Nav channel activator in fluorescence based assays (Deuis et al., 2015; Vetter et al., 2012a; Vickery et al., 2004). In rodent dorsal root ganglion neurons, veratridine causes persistent current in TTXs channels but not in TTXr channels, suggesting differential effects on Nav1.8 and Nav1.9 (Farrag et al., 2008).

Veratridine shares an overlapping but non-identical binding site with batrachotoxin, as the poison dart species *Phyllobates terribilis* is sensitive to veratridine, but not batrachotoxin (Daly et al., 1980). In addition, single point mutations of amino acids located on S6 in DI (I433, N434, L437) and S6 in DIV (F1579, N1584) cause rNav1.4 to become insensitive to the inactivation-modulating effects of veratridine (Figs. 4 and 8), although all of the mutants remained sensitive to the inhibitory effects of veratridine on peak current (Wang et al., 2000a). The reason for this disparity remains unclear, but suggests that the persistent current and reduction in peak current are caused by two distinct channel interactions. The most likely explanation is that peak current is reduced after binding of veratridine due to partial occlusion of the ion permeation pathway, while gating/inactivation cannot be modulated anymore in these mutants due to removal of the essential gating-modifying interactions.

Aconitine is a steroid-derived alkaloid found in the *Aconitum napellus* (monkshood) plant. Like batrachotoxin and grayanotoxin, aconitine preferentially binds to the open channel state, requiring repetitive depolarising pulses to see a functional effect under voltage-clamp conditions (Rao and Sikdar, 2000). Aconitine causes

a hyperpolarising shift in the voltage-dependence of activation, inhibits peak current, slows the rate of fast inactivation (although aconitine-modified channels can inactivate completely), and causes a reduction in ion selectivity (Rao and Sikdar, 2000; Wright, 2002). Aconitine has similar pharmacological effects at Nav1.2, Nav1.4 and Nav1.5, although a small persistent current is seen at Nav1.5 only (Rao and Sikdar, 2000; Wright, 2002). Activity at other Nav channel subtypes in heterologous expression systems remains to be determined. The binding site is assumed to overlap with other site 2 toxins, based on similar chemical structures and mechanism of action, although this remains to be experimentally confirmed.

4.1.2. Spider-venom derived Nav channel activators

Spider venom-derived peptides have diverse pharmacological effects at Nav channels as they can interact with different voltage sensor(s). The resultant effects including inhibition of channel opening, a shift the voltage-dependence of activation, and/or a delay in fast inactivation. In general, spider venom-derived peptides that interact exclusively with DIV delay fast inactivation, while peptides that interact with DI-III result in inhibition of channel opening and/or shifts in the voltage-dependence of activation (Bosmans et al., 2008). Accordingly, spider venom-derived peptides are named using Greek symbols that denote their pharmacological activity, with the prefix μ-, β-, or δ-used to described toxins that inhibit Nav channels, toxins that shift the voltage dependence of activation or toxins that delay fast inactivation, respectively, followed by taxonomic family, then a genus/species and numerical descriptor (King et al., 2008). For consistency, the nomenclature utilised in ArachnoServer will be used to identify spider venom-derived peptides in this review and peptides will be grouped into

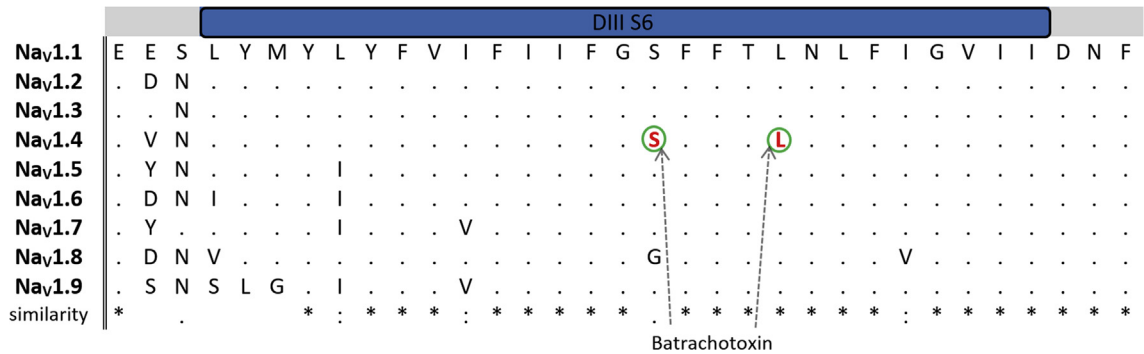


Fig. 7. Amino acid sequence alignment of human Nav isoforms for DIII S6. Multiple sequence alignment of domain III segment 6 and adjacent residues was performed as stated in Fig. 4. This segment contains important residues for batrachotoxin binding.

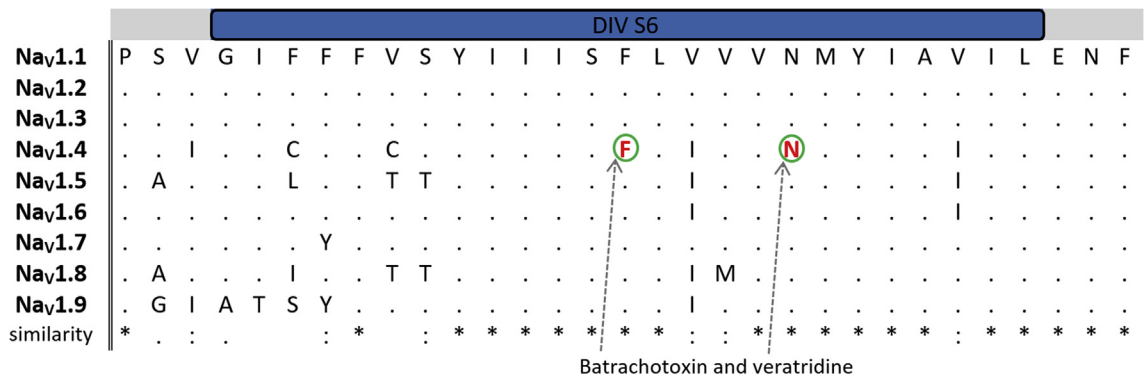


Fig. 8. Amino acid sequence alignment of human Nav isoforms for DIV S6. Multiple sequence alignment of domain IV segment 6 and adjacent residues was performed as stated in Fig. 4. DIV S6 contains residues that are important for batrachotoxin and veratridine binding.

Family's 1–12 based on sequence homology as proposed by Klint et al. (Herzig et al., 2011; Klint et al., 2012).

Venom from two Australian funnel-web spider species, *Atrax robustus* and *Hadronyche versuta*, and the Australian eastern mouse spider *Missulena bradleyi* are the source of four homologous Nav channel activator toxins consisting of 42 amino acid residues belonging to Family 4, named δ -hexatoxin-Ar1a (formally robustoxin), δ -hexatoxin-Hv1a (formally versutoxin), δ -hexatoxin-Hv1b and δ -actinopoditoxin-Mb1a. In dissociated rat DRG neurons, these toxins potently inhibit fast inactivation of TTX-sensitive channels, with no effect on TTX-resistant channels (Alewood et al., 2003; Gunning et al., 2003; Nicholson et al., 1996, 1998; Szeto et al., 2000). Consistent with this activity, δ -hexatoxin-Ar1a and δ -hexatoxin-Hv1a competitively inhibit binding of the α -scorpion toxin Lqh II in rat brain synaptosomes, indicating they interact with DIV (Little et al., 1998). In addition, these toxins also cause a hyperpolarizing shift in the voltage-dependence of activation and voltage-dependence of steady-state inactivation (Alewood et al., 2003; Gunning et al., 2003; Nicholson et al., 1996, 1998; Szeto et al., 2000), suggesting additional interactions with DII, although the exact binding site(s) of this family remains to be elucidated. The only member of Family 4 to be characterised at Nav1.1–1.8 is δ -hexatoxin-Mg1a (formally Magi 4) from the Japanese funnel-web spider *Macrothele gigas*. At a concentration of 5 μ M, δ -hexatoxin-Mg1a delays fast inactivation, with a large effect at Nav1.1 and Nav1.6, a comparatively smaller effect at Nav1.2, Nav1.3 and Nav1.7, and no effect at Nav1.4, Nav1.5 and Nav1.8 (Yamaji et al., 2009). Like members of Family 4 isolated from Australian spiders, δ -hexatoxin-Mg1a also shifts the voltage-dependence of activation to more hyperpolarized potentials.

Another Nav channel activator isolated from the Japanese funnel-web spider *Macrothele gigas* is β -hexatoxin-Mg1a (formally Magi 5; also identical primary sequence to β -hexatoxin-Mr1a from *Macrothele raveni*), a 29-residue peptide that has little sequence homology to δ -hexatoxin-Mg1a and belongs to Family 5 (Corzo et al., 2007; Zeng et al., 2003). β -hexatoxin-Mg1a has only been characterised at Nav1.2, where it shifts the voltage-dependence of activation to more hyperpolarized potentials, with no effect on the rate of fast inactivation (Corzo et al., 2007). Consistent with this activity, in rat brain synaptosomes β -hexatoxin-Mg1a competitively inhibits binding of the β -scorpion CssiV, but not the α -scorpion toxin Lqh II, indicating it interacts with the DII voltage sensing domain (Corzo et al., 2003).

Over 90 peptides have been sequenced from the venom of the Chinese tarantula species *Chilobrachys guangxiensis* (also known as *Chilobrachys jingzhao*). While the pharmacological activity of most of these peptides remains to be confirmed, several have been

identified Nav channel activators (Chen et al., 2008). δ -theraphotoxin-Cg1a (formerly Jingzhaotoxin-I) and δ -theraphotoxin-Cg3a (formerly Jingzhaotoxin-II) consist of 33 and 32 amino acid residues, respectively, with ~50% sequence homology and belong to Family 7 (Wang et al., 2008b; Xiao et al., 2005). Both δ -theraphotoxin-Cg1a and δ -theraphotoxin-Cg3a have similar pharmacological activity and delay inactivation of Nav channels, with little to no effect on the voltage-dependence of activation or steady state inactivation (Huang et al., 2015; Tao et al., 2016; Wang et al., 2008b; Xiao et al., 2005). Consistent with effects on inactivation, site directed mutagenesis of D1609A on hNav1.5, which is located on the DIVS3-S4 linker (Fig. 9), causes both δ -theraphotoxin-Cg1a and δ -theraphotoxin-Cg3a to lose activity, confirming that these peptides interact with the voltage-sensing domain of DIV (Huang et al., 2015; Tao et al., 2016). Interestingly, the selectivity profiles of these peptides differs from each other, with δ -theraphotoxin-Cg3a most potentially delaying inactivation at Nav1.5 (EC₅₀ 125 nM), with minor effects on Nav1.3, Nav1.4 and Nav1.7 at 5 μ M, whilst δ -theraphotoxin-Cg1a is comparatively non-selective, delaying inactivation at Nav1.2, Nav1.3, Nav1.4, Nav1.5 and Nav1.7 with similar potency (EC₅₀ 335–870 nM) (Huang et al., 2015; Tao et al., 2016). At higher concentrations, δ -theraphotoxin-Cg1a also inhibits Kv2.1 (Tao et al., 2013).

δ -theraphotoxin-Cg2a (formerly Jingzhaotoxin-IV) is a 34-residue peptide also from the venom of *Chilobrachys guangxiensis*, but that shares sequence homology to Family 2. It delays inactivation of TTX-sensitive channels in rat DRG neurons, but unlike the other spider-venom derived Nav channel activators described so far, causes a depolarising shift in the voltage-dependence of activation, as well as a hyperpolarising shift in the voltage-dependence of inactivation (Wang et al., 2008a). The binding site of δ -theraphotoxin-Cg2a remains to be elucidated but based on its pharmacological activity, is likely to involve interactions with DII and DIV. While interactions with other domains cannot be ruled out, DIII interactions are yet to be described for spider toxins. Full Nav channel selectivity is not available for δ -theraphotoxin-Cg2a.

Many members of Family 2 inhibit Kv channels (denoted by the prefix κ). Despite being originally identified as Kv inhibitors, an increasing number of peptides from Family 2 are being found to act on Nav channels. One notable example is δ -theraphotoxin-Hm1a, which was originally described as a Kv2 and Kv4 inhibitor. Interestingly, Hm1a selectively delays inactivation of Nav1.1 (EC₅₀ 38 nM), with no effect on Nav1.2–1.8 at a concentration of 100 nM (Osteen et al., 2016). Construction of Kv2.1/hNav1.1 chimeric channels, where the S3-S4 linker from each domain of hNav1.1 is inserted into the homotetrameric Kv2.1 channel, has identified that δ -theraphotoxin-Hm1a binds exclusively to DIV, consistent with its

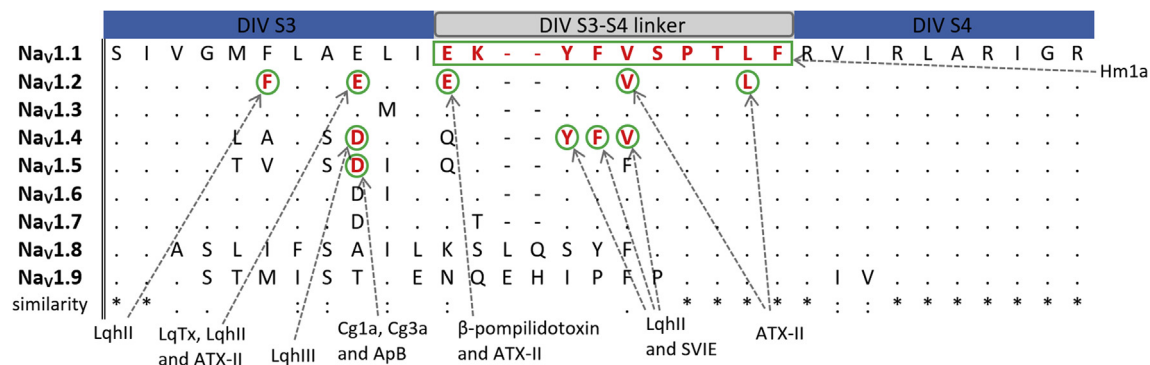


Fig. 9. Amino acid sequence alignment of human Nav isoforms for DIV VSD S3-S4 linker. Multiple sequence alignment of domain IV S3-S4 linker and adjacent residues was performed as stated in Fig. 4. The shown region contains important residues for interactions with the δ -spider toxins Cg1a, Cg3a and Hm1a β -pompilidotoxin, the α -scorpion toxins LqTx, LqhII and LqhIII, the δ -conotoxin SVIE, and the sea anemone toxins ATX-II and ApB.

pharmacological activity (Figs. 9 and 10) (Osteen et al., 2016).

4.1.3. Wasp venom-derived Nav channel activators

Two 13 amino acid residue peptides, named α -pompilidotoxin and β -pompilidotoxin, isolated from the venom of the solitary wasps *Anoplius samariensis* and *Batozonellus maculifrons*, respectively, are known to activate Nav channels (Konno et al., 1998). They are highly homologous, differing by only one amino acid, with most work carried out using β -pompilidotoxin, as it is 5 times more potent than α -pompilidotoxin (Konno et al., 2000).

β -pompilidotoxin has some selectivity between Nav channels subtypes: it delays fast inactivation at Nav1.1, Nav1.2 (EC₅₀ 21 μ M), Nav1.3 (EC₅₀ 99 μ M), and Nav1.7, inhibits fast inactivation at Nav1.6 (EC₅₀ 30 μ M), and has no effect on the fast inactivation kinetics of Nav1.4 and Nav1.5 (up to 140 μ M) (Schiavon et al., 2010). Using a range of rNav1.2/rNav1.5 channel chimeras, the binding site of β -pompilidotoxin was identified as the S3-S4 linker of DIV, consistent with its pharmacological effects on delaying fast inactivation (Kinoshita et al., 2001). The amino acid E1616 in the S3-S4 linker of DIV rNav1.2, which is conserved in Nav1.1, Nav1.2, Nav1.3, Nav1.6, Nav1.7 but not in Nav1.4 or Nav1.5, is essential for β -pompilidotoxin binding, and accounts for this Nav channel selectivity profile (Fig. 9) (Kinoshita et al., 2001).

4.1.4. Snake venom-derived Nav channel activators

Three-finger toxins (3FTxs) are a large family of snake venom-derived toxins with a conserved structural fold but diverse pharmacology, which includes activity at nicotinic and muscarinic acetylcholine receptors, acetylcholinesterase and L-type voltage-gated calcium channels (Kini and Doley, 2010). δ -callitoxin, a 57-amino acid residue peptide isolated from the long-glanded blue coral snake (*Calliophis bivirgatus*), is the first 3FTx to be identified as a Nav channel activator (Yang et al., 2016). At Nav1.4, δ -callitoxin (200 nM) causes a small hyperpolarising shift in the voltage-dependence of activation and delays fast inactivation, as well as causing a persistent current (Yang et al., 2016). Activity and selectivity at other Nav channel subtypes remains to be determined. Although the binding site of δ -callitoxin has not yet been defined, based on its pharmacological activity, it likely involves interactions with residues of DIV.

4.1.5. Scorpion venom Nav channel activators

Scorpion neurotoxins that act on Nav channels are broadly categorised as α or β toxins – both of which contain 60–80 amino acids linked by four disulphide bridges – based on their effects on the electrophysiological properties of Nav channels. This functional

distinction is a reflection of distinct binding sites on Nav channels (Bosmans and Tytgat, 2007; Pedraza Escalona and Possani, 2013), with α toxins binding to site 3 to cause a delay in fast inactivation (Gordon et al., 1996). In contrast, β -scorpion toxins bind to site 4 and alter the threshold for activation, allowing channel opening at hyperpolarized potentials whilst also reducing peak current (Couraud et al., 1982). Thus, both α - and β -scorpion toxins are appropriately described as Nav channels activators albeit with a distinct mechanism of action. Scorpion neurotoxins have proved useful in characterising Nav channel binding sites and function, although it is important to note that with increased understanding of Nav channel structure, a more complex picture of toxin binding has emerged. For brevity, here we discuss only those toxins that are only active at mammalian Nav channels and also display some selectivity across the isoforms (for comprehensive review of scorpion toxins see (Housley et al., 2016)).

Structural and functional assessment of α -scorpion toxins has led to a model of toxin binding that impairs the movement of the VSD of domain IV (Cestele and Catterall, 2000), which as detailed above, is critical for inactivation but not activation gating. Experimentally, classical α scorpion binding to site 3 manifests as a delay in inactivation without major changes in channel activation (Bosmans and Tytgat, 2007). Furthermore, α -scorpion toxins, which can in addition be classified into α or α -like based on functional effects and target species, may be “pulsed off” with strong depolarizations (Bosmans and Tytgat, 2007; Campos et al., 2004; Catterall, 1977).

The classical old world toxin, AaHII, isolated from *Androctonus australis*, is a prototypical α -scorpion toxin. The selectivity of AaHII has not been reported, however it readily binds to rat brain synaptosomes presumably expressing Nav1.2 (Legros et al., 2005).

Venom from *Leiurus quinquestriatus hebraeus* contains α -toxins LqhII and LqhIII that are highly toxic to mammals (Sautiere et al., 1998). LqhII and LqhIII modulate Nav1.4 and Nav1.5 by removing fast inactivation as described above (Chen et al., 2000; Chen and Heinemann, 2001). Additionally, LqhII and III enhance slow inactivation of Nav1.5 as some of the first α -scorpion toxins to cause this phenomenon (Chen and Heinemann, 2001). LqhIII also alters the kinetics of hNav1.2 and hNav1.7 and is known as an α -like toxin, binding to both mammalian and insect Nav channels, but not rat synaptosomes (Chen et al., 2002; Gilles et al., 2000). LqTx (*Leiurus quinquestriatus*) highlighted the crucial role for E1613 (rNav1.2) in site 3 toxin binding (Rogers et al., 1996), which was further investigated with LqhII and LqhIII. Interestingly, D1428 of rNav1.4 (analogous to E1613 rNav1.2) is critical to LqhIII but not LqhII binding (Leipold et al., 2004). LqhII relies on residues Y1433, F1434

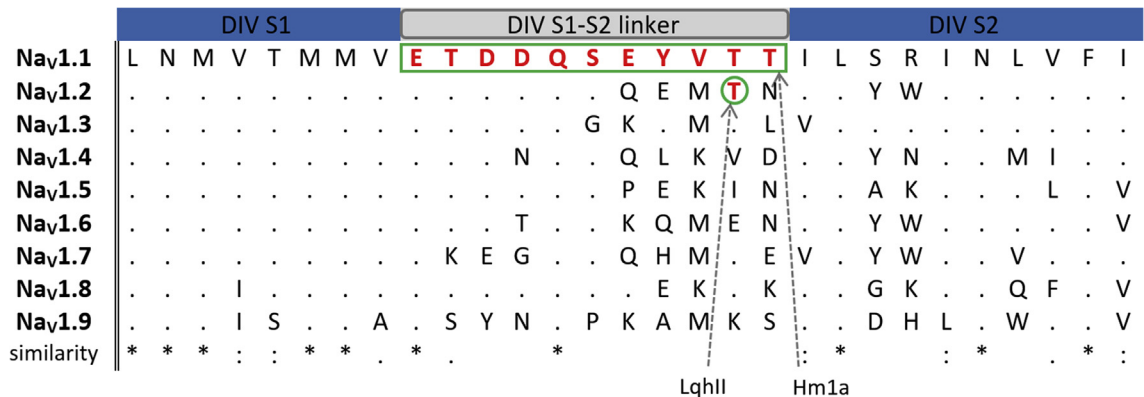


Fig. 10. Amino acid sequence alignment of human Nav isoforms for DIV S1–S2 linker. Multiple sequence alignment of domain IV S1–S2 linker and adjacent residues was performed as stated in Fig. 4. The DIV S1–S2 linker contains important residues for LqhII and Hm1a binding.

and V1435 of DIV S3–S4 linker (rNav1.4) as site directed mutagenesis causes loss of activity (Fig. 9) (Leipold et al., 2005). Furthermore, a secondary interaction site for LqhII is present in the DI S5–S6 linker (Wang et al., 2011). In particular, an alanine scan of DI highlighted T393 (rNav1.2) as a crucial binding partner of LqhII (Fig. 11).

The α -scorpion toxin OD1, isolated from the Iranian scorpion *Odonthobuthus doriae* was initially described to interact with the insect Nav channel isoform para1pE. However, OD1 is also an activator of hNav1.7, hNav1.4 and hNav1.6 as measured by fluorescence based assays and electrophysiology (Durek et al., 2013a). At low nanomolar concentrations, OD1 significantly impairs inactivation of Nav1.7 (Maertens et al., 2006b; Motin et al., 2016), while at higher concentrations (300 nM), OD1 also causes a hyperpolarising shift in the voltage dependence of activation, considerably increases peak current values and induces a large persistent current (Deuis et al., 2016b).

Ts2, from *T. serrulatus*, prevents inactivation of rat Nav1.2, Nav1.3, Nav1.5, Nav1.6 and Nav1.7 (Cologna et al., 2012). However, Ts2 also selectively shifts the voltage dependence of activation of rNav1.3 (Cologna et al., 2012). Accordingly, although Ts2 is currently classified as an α scorpion toxin, this specific activation of rNav1.3 is similar to β -scorpion Tf2 discussed below. Thus, the mixed pharmacological effects of both OD1 and Ts2 blur the line between α - and β -scorpion toxins, inviting a revision of the classification these Nav channel activators.

MeuNaTx α s are a large family of thirteen α -scorpion toxins isolated from the scorpion *Mesobuthus eupeus*. Of these, the pharmacological activity of MeuNaTx α -1, 2 and 5 has been comprehensively characterised. MeuNaTx α -1 modulates Nav1.2, Nav1.3, Nav1.6 and Nav1.7, MeuNaTx α -2 affects Nav1.4 only, whereas MeuNaTx α -5 is the least selective modulating Nav1.3–1.7 (Zhu et al., 2012). These peptides also potently slow inactivation in the insect channel counterpart DmNav1. MeuNaTx α -12 and MeuNaTx α -13 also preferentially affect Nav1.1 (Zhu et al., 2013).

Bot IX, from *Buthus occitanus tunetanus* is a recently reported α -like toxin, inhibiting fast inactivation at both insect Nav and rNav1.2 (Martin-Eauclaire et al., 2016). However, Bot IX also competed with AaIII binding, the first α -like toxin described to do so (Martin-Eauclaire et al., 2016). Full selectivity of Bot IX at mammalian Nav channel isoforms is unknown.

Scorpions of the genus *Centruroides* produce several peptides that were critical for early understanding of the mechanics of the VSD in Nav channels (Cestele et al., 1998). Specifically, the class of β -scorpion toxins was based on characterisation of CssIV, from *C. suffusus*, which shifts the voltage dependence of activation of Nav1.1, 1.2 and 1.6 (Jover, 1980; Jover et al., 1980; Karbat et al., 2010;

Schiavon et al., 2006). This activity is explained through a voltage sensor trapping mechanism whereby the peptide binds to the activated position of the DII VSD (both S1–S2 and S3–S4 extracellular loops), locking it in an activated position (Cestele et al., 1998; Pedraza Escalona and Possani, 2013; Zhang et al., 2011; Cestele, 2006). Regions crucial for CssIV binding are found in DI S5–S6, DII S1–S2 and S3–S4 linkers and various residues located in the DIII S5–S6 linker (Cestele et al., 1998). This includes P782 (DII S1–S2) and G845 (DII S3–S4) in rNav1.2, which when substituted for asparagine result in reduced CssIV binding and in the case of G845, a loss of activity. Within the DIII S5–S6 linker, mutations of N1436, E1438, L1439 and D1445 (rNav1.2) cause changes in CssIV binding and voltage-sensor trapping activity (Figs. 6 and 11–13) (Zhang et al., 2011).

Similarly, the β scorpion toxin Cn2, originally purified as fraction II-10 from the crude venom of *Centruroides noxius* Hoffmann, is particularly notable for its high selectivity for hNav1.6 (Schiavon et al., 2006). Cn2 causes a robust hyperpolarising shift in activation permitting Nav1.6 opening at usually prohibitive membrane potentials, albeit this effect is only observed during protocols that includes a prepulse to allow toxin binding (Cestele et al., 1998; Schiavon et al., 2006).

Intriguingly, Cn2 also enhances resurgent currents in Purkinje neurons (Schiavon et al., 2006), where non-conducting (inactivated) Nav channels are seen to reopen (resurge) in response to mild repolarization steps from positive potentials (Raman and Bean, 1997). This resurgent current is reduced by 90% in Nav1.6 KO mice leading to the identification of Nav1.6 as the primary channel responsible for carrying this current (Raman et al., 1997). Resurgent currents are thought to allow high frequency firing in neurons and may be involved in clinical disease states such as paroxysmal extreme pain disorder and chemotherapy induced cold allodynia (Jarecki et al., 2010; Sittl et al., 2012). Interestingly, although CssIV has 83% sequence homology with Cn2 and is capable of binding Nav1.6, it does not produce a resurgent current (Schiavon et al., 2006). Thus, Cn2 may find future application as a probe to understand the intricacies of resurgent current *in vitro* and *in vivo*.

The genus of *Tityus* scorpions, whose habitat ranges from Central to South America, has provided several β -scorpion toxins with interesting pharmacological properties. For example, Ts1 from *T. serrulatus* shifts the voltage dependence of activation of Nav1.2, 1.4 and Nav1.6 and also decreases peak current of Nav1.6, and to a lesser extent Nav1.4 (Peigneur et al., 2015a). Interestingly, this toxin also exerts a full block of Nav1.5 without changes in activation or inactivation (Peigneur et al., 2015a). In contrast, Tz1 from *T. zulianus* also modulates the activity of Nav1.4 > Nav1.6 > Nav1.2 but not

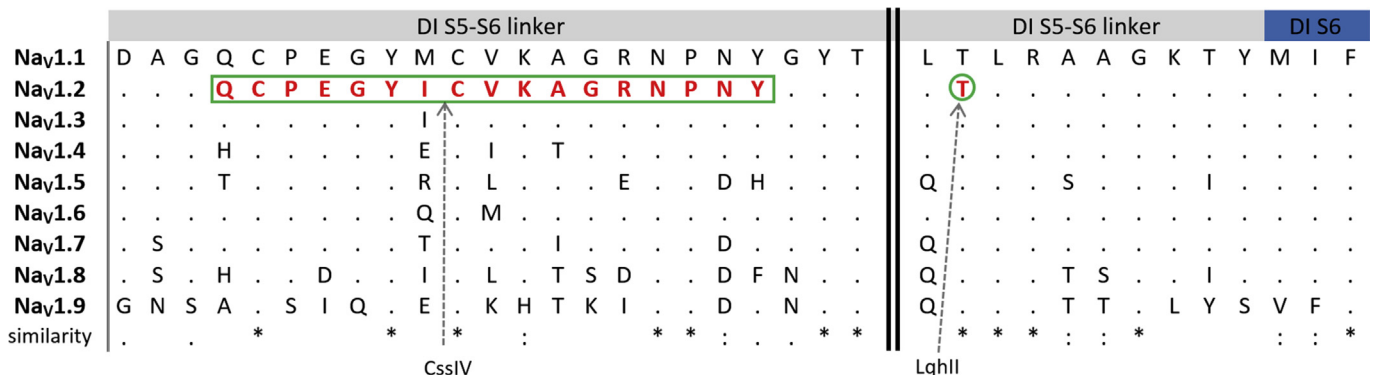


Fig. 11. Amino acid sequence alignment of human Nav isoforms for DI S5–S6 linker. Multiple sequence alignment of part of the domain I S5–S6 linker was performed as stated in Fig. 4. The vertical lines indicate a break in the sequence. This region contains important residues for CssIV and LqhII binding.

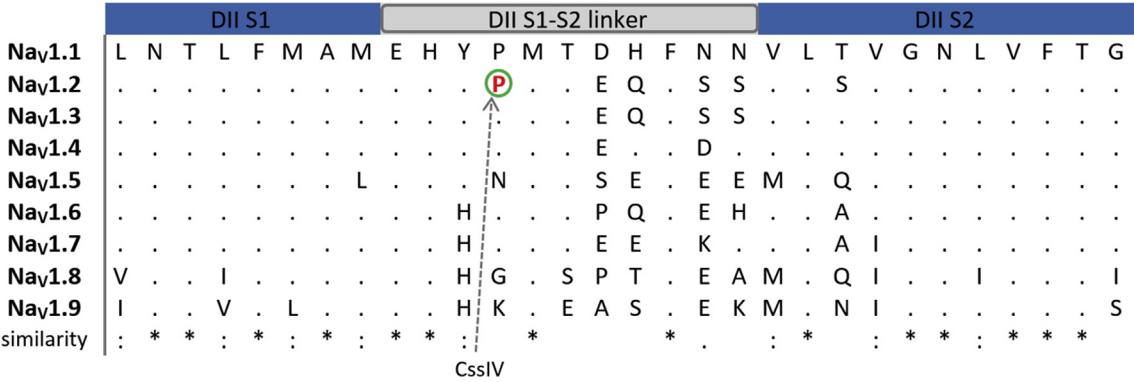


Fig. 12. Amino acid sequence alignment of human Nav isoforms for DII S1–S2 linker. Multiple sequence alignment of domain II S1–S2 linker and adjacent residues was performed as stated in Fig. 4. The DII S1–S2 linker contains a proline important for CssIV binding.

Nav1.5 and Nav1.7 (Leipold et al., 2006). Nav channel residues involved in Tz1 activity are shown in Figs. 6 and 13. Highlighting the pharmacological diversity of scorpion toxins, Tf2 from *T. fasciolatus* selectively causes the opening hNav1.3 at hyperpolarized membrane potentials (Camargos et al., 2015) with no effect (activation or inactivation) on other Nav channel isoforms. This highly specific activation of a singular Nav channel isoform may be useful as a tool to delineate the physiological and pathological role of Nav1.3 (Hains et al., 2003, 2005; He et al., 2010).

4.1.6. Cone snail venom derived Nav channel activators

Marine snails of the genus *Conus* have evolved a complex cocktail of venom peptides (conotoxins) for both predatory and defensive behaviour (Dutertre et al., 2013; Lewis et al., 2012). These small peptides are universally cysteine-rich and folded with multiple disulfide bridges (Han et al., 2008; Lewis et al., 2012; Vetter and Lewis, 2012). Conotoxins of the μ -, μ O-, δ - and ι families target Nav channels via a variety of mechanisms (Ekberg et al., 2008; Green and Olivera, 2016). However, as μ - and μ O-conotoxins are Nav channel inhibitors this review will focus on δ - and ι -conotoxins.

δ -Conotoxins inhibit fast inactivation and modulate the voltage-dependence of activation of Nav channels (Ekberg et al., 2008; Stevens et al., 2011). The result of this hyperpolarizing shift is prolonged channel opening and thus, persistent firing of excitable cells (Ekberg et al., 2008). In-depth information regarding structure-function relationship between δ -conotoxins and their Nav channel targets is sparse owing in part to challenges in

chemical synthesis and the extreme hydrophobicity of these peptides. Thus, the exact binding site of the δ -conotoxins remains a point of contention. As their pharmacological effects are similar to those of the α -scorpion toxins, site 3 was proposed as a binding site; however, δ -conotoxins do not fully compete with site 3 toxins (Fainzilber et al., 1994; Leipold et al., 2005). Furthermore, site-directed mutagenesis of this region failed to alter all biophysical properties associated with δ -conotoxin binding (Leipold et al., 2005). SVIE (*Conus striatus*) interacts with a hydrophobic triad of the DIV S3-S4 linker, Y1433–F1434–V1435 (rNav1.4) (Fig. 9) (Leipold et al., 2005). Thus, a new binding site (site 6) was suggested, although it remains poorly defined.

While δ -conotoxins are commonly found in vermivorous or piscivorous *Conidae* and activity at Nav channel homologues of worms and fish might contribute to the use of these peptides in prey capture, they also affect mammalian Nav channels. For example, δ -EVIA, isolated from *C. ermineus*, inhibits fast inactivation of rNav1.2, rNav1.3 and rNav1.6 without altering rNav1.4 and hNav1.5 (Barbier et al., 2004; Volpon et al., 2004). In addition, δ -Am2766 from *C. amadis* inhibits inactivation of rNav1.2, albeit activity at other subtypes is unknown (Sarma et al., 2005). Interestingly, δ -SuVIA, isolated from the vermivorous *C. suturatus*, is active at low nanomolar concentrations at hNav1.3, hNav1.4, hNav1.6 and hNav1.7 (Jin et al., 2015). This defensive conotoxin causes a hyperpolarizing shift in activation potential at hNav1.7 but fails to measurably modify inactivation kinetics (Jin et al., 2015). This is at clear odds with other δ -conotoxins studied from fish-hunting *Conus* and may suggest a different class of conotoxin altogether.

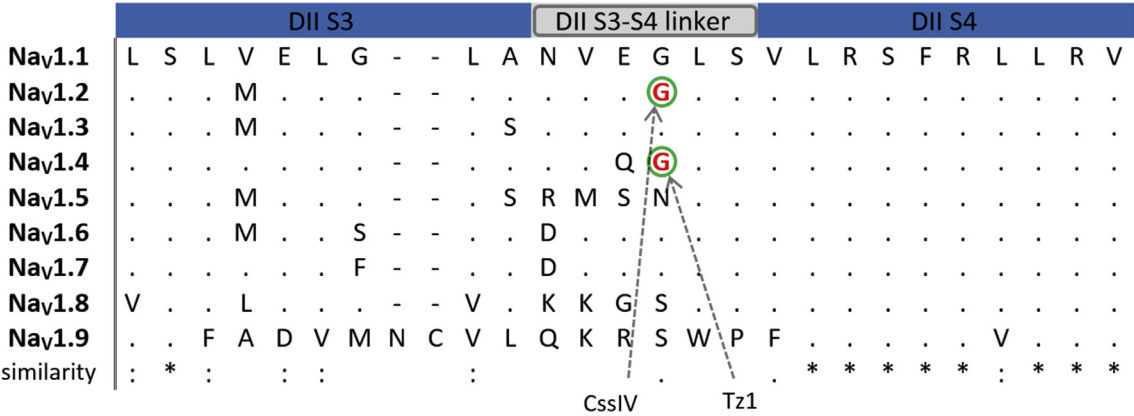


Fig. 13. Amino acid sequence alignment of human Nav isoforms for DII S3–S4 linker. Multiple sequence alignment of domain II S3–S4 linker and adjacent residues was performed as stated in Fig. 4. The DII S3–S4 linker contains important residues for CssIV and Tz1 binding.

The superfamily of I1 conotoxins was initially identified in the venom of *C. radiatus* (Jimenez et al., 2003). These peptides contain a unique cysteine backbone structure with four disulfide bonds (Buczek et al., 2007). ι -conotoxin RXIA shifts the voltage-dependence of activation, without altering inactivation, most potently at rNa_v1.6 (EC₅₀ 2 μ M) followed by rNa_v1.2 and rNa_v1.7 (Fiedler et al., 2008). As ι -RXIA had no effect on other subtypes tested, the ι -conotoxins may provide an additional source of selective agonist peptides.

Additionally, tetrodotoxin-sensitive current in DRGs were enhanced by M-superfamily toxin, LtIIIA isolated from *C. litteratus* (Wang et al., 2009). Given that DRGs express a range of Na_v channel isoforms, the selectivity of this peptide is unknown.

4.1.7. Sea anemone derived Na_v channel activators

Sea anemone toxins that associate with Na_v channels are categorised into three groups: Type I, II and III (Frazao et al., 2012; Moran et al., 2009; Norton, 1991). However, in recent years questions about the validity of these broad classifications have arisen as additional peptides are discovered and characterised that diverge from the stringent structural and functional properties proposed originally (Ishida et al., 1997; Moran et al., 2009). Sea anemone toxins bind to site 3 and produce a delay in inactivation at low concentrations as well as a large persistent current at high concentrations (Catterall and Beress, 1978; Moran et al., 2007; Smith and Blumenthal, 2007). ATX-II binds the DIV S3–S4 linker, in particular E1613 (rNa_v1.2) where charge inversion at this point results in no binding (Rogers et al., 1996). This is direct evidence for the overlap between ATX-II and other site 3 toxins including LqTx described above. Furthermore E1616, V1620 and L1624 alanine substitutions resulted in lower binding affinity of ATX-II but not LqTx (Fig. 9). Altering the corresponding amino acid of rNa_v1.5, D1612, decreased binding affinity for ApB (*Anthopleura xanthogrammica*) (Benzinger et al., 1998). Taken together, this region of the DIV S3–S4 loop is indeed central to sea anemone toxin binding. The Na_v channel subtype selectivity for mammalian isoforms has been systematically assessed for a limited number of peptides, including ATX-II, AFT-II and Bc-III isolated from *Anemonia sulcata*, *Anthopleura fuscoviridi* and *Bunodosoma granulifera*, respectively (Bosmans et al., 2002; Oliveira et al., 2004). Although these toxins display differential effects at Na_v1.1–1.6, none are truly selective activators (Table 1) (Oliveira et al., 2004).

Other Na_v channel activators present in sea anemone venom include CGTX-II and δ -AITX-Bcg1a, from *Bunodosoma cangicum*. CGTX-II interacts with hNa_v1.5 over Na_v1.6 with minor effects at Na_v1.1, 1.2, 1.3, 1.4 and 1.7 (Zaharenko et al., 2012). Interestingly, CGTX-II causes a depolarising shift in the inactivation curve of only hNa_v1.5, whereas the effect on hNa_v1.6 includes a large non-inactivating window current (Zaharenko et al., 2012). While the molecular basis of these differential effects remains unclear to date, these pharmacological effects may provide additional insight into Na_v channel gating mechanisms and structural motifs contributing to effects on channel activation and inactivation.

4.1.8. Dinoflagellate and bacterial Na_v channel activators

Na_v channel activators of marine dinoflagellate origins are the cause of several seafood related food poisoning diseases including ciguatera and “red-tide” shellfish poisoning (Gillespie et al., 1986; Nakanishi, 1985; Swift and Swift, 1993). The causative cyclic polyether toxins, the ciguatoxins (CTX) and brevetoxins (PbTx), are produced by dinoflagellates of the *Gambierdiscus* and *Karenia* families, respectively (Bidard et al., 1984). The pharmacological effects of these site 5 toxins on Na_v channels are diverse and include hyperpolarising shifts in activation, a decrease in peak current, as well as effects on the voltage-dependence and time

course of channel inactivation (Inserra et al., 2017; Lombet et al., 1987; Trainer et al., 1994). Critical residues for site 5 binding are less well defined, although photoaffinity labelling studies with PbTx highlight DI S6 and most of the DIV S5–S6 linker, including the pore loop with the residues that form part of the sodium selectivity filter (Figs. 4 and 14) (Trainer et al., 1994).

PbTx's are classified as either A or B depending on their backbone structure. The extensively studied PbTx-3 shifted the voltage-dependence of activation of tetrodotoxin-sensitive channels expressed in nodose ganglia (Jeglitsch et al., 1998), inhibited Na_v channel inactivation, and increased channel opening time in single channel recordings (Purkerson et al., 1999). Although PbTx-3 showed some selectivity for the skeletal muscle isoform (Na_v1.4) over the cardiac (Na_v1.5) and rat brain (Na_v1.2) isoforms in a sodium dependent cytotoxicity assays (Bottein Dechraoui and Ramsdell, 2003), selectivity across the full panel of Na_v channels is unknown, and selectivity of other PbTx's has thus far not been studied.

The ciguatoxins are Na_v channel activator toxins that accumulate in fish through the marine food chain and cause the seafood disease ciguatera in humans. Of the several structurally related congeners, denoted as Pacific, Indian and Caribbean ciguatoxins depending on their origin, P-CTX-1 (Pacific ciguatoxin 1) is the most potent, and the isoform responsible for most symptoms in the Pacific (Lewis, 2006). Although differential effects on tetrodotoxin-resistant and -sensitive isoforms expressed in sensory neurons – including shifts in the voltage-dependence of activation, a decrease in peak current, and effects on recovery from inactivation – were reported several years ago, the relative Na_v channel subtype selectivity of P-CTX-1 has remained unexplored until relatively recently (Strachan et al., 1999). In SH-SY5Y cells, P-CTX-1-induced effects were mediated predominantly through Na_v1.3 (Vetter et al., 2012b), consistent with the induction of a small persistent current at this, but not other, Na_v channel isoforms (Inserra et al., 2017). While P-CTX-1 is relatively non-selective for Na_v1.1–Na_v1.9 overall, as evidenced by a shift in the voltage-dependence of activation at all isoforms, effects on fast inactivation only occurred at Na_v1.2, Na_v1.3 and Na_v1.9.

Antillatoxin is a Na_v channel activator isolated from the cyanobacteria *Lyngbya majuscula* (Orjala et al., 1995) and causes Na⁺ influx via rNa_v1.2, 1.4 and 1.5 (Cao et al., 2010). Additionally, antillatoxin causes changes in membrane potential resulting from hNa_v1.7 activation (Zhao et al., 2016). While there is evidence for allosteric coupling with site 2 and 5, the exact binding site remains unspecified (Cao et al., 2010; Li et al., 2001). Interestingly, most studies assessing the pharmacology of antillatoxin were conducted using fluorescence-based assays, and electrophysiological characterisation of its effects on Na_v channel gating, particularly at different isoforms, is sparse.

5. Na_v channel activators as tool compounds

5.1. Na_v channel assays

Na_v channels are key therapeutic targets for pain, epilepsy and cardiac arrhythmias; therefore several high-throughput assays have been developed to screen and profile the activity of Na_v channel modulators, including fluorescence-based assays and automated patch-clamp electrophysiology (Dunlop et al., 2008; Yu et al., 2016). Fluorescence-based assays provide an indirect measure of Na_v channel function by detecting changes in membrane potential or intracellular sodium ion concentration, with several fluorescent dyes commercially available (Yu et al., 2016). Whilst the results from fluorescence-based assays do not always correlate well with patch-clamp electrophysiology, which remains the standard

[illegible]

Fig. 14. Amino acid sequence alignment of the domain IV S5–S6 linker. Multiple sequence alignment of part of the domain IV S5–S6 linker. The shaded conserved alanine and aspartic acid residues are part of the channels sodium selectivity filter in the P-loop of DIV. This region contains important residues for interactions with brevetoxin and ciguatoxin.

for measuring Nav channel function, they are comparatively cheaper, higher throughput, and do not require significant expertise to perform (Dunlop et al., 2008; Yu et al., 2016).

To activate Na_v channels in fluorescence-based assays, the alkaloid veratridine or the pyrethroid deltamethrin are commonly used. Veratridine induces a robust fluorescent response with membrane potential dyes in cells heterologously expressing Na_v1.1–1.7, but has negligible effects on Na_v1.8 (Deuis et al., 2015; Vickery et al., 2004). In contrast, deltamethrin has a limited effect on the membrane potential of cells heterologously expressing TTXs Na_v channels, but causes a robust fluorescent response in cells heterologously expressing Na_v1.8 (Deuis et al., 2016a; Vickery et al., 2004). As Na_v channel activators used in fluorescence-based assays likely stabilise different channel conformations, it is unclear if these assays exhibit bias towards detection of Na_v channel modulators with a specific mechanism of action and/or binding site.

5.2. *Nav* channel target engagement in vivo

A key step in the development of any pharmaceutical is to demonstrate target engagement and efficacy *in vivo*. For analgesics, assessment of target-engagement *in vivo* is challenging, as most commonly used pre-clinical rodent models of pain, which are based on quantification of pain responses induced by local injection of formalin, carrageenan or Complete Freund's Adjuvant, have a multifactorial pathophysiology. Demonstrating target engagement *in vivo* ensures a sufficient concentration of an analgesic is reached at the site of the intended target, providing crucial information to guide dosing for *in vivo* efficacy studies.

Nav1.7 is of particular interest as an analgesic target because loss-of-function mutations in *SCN9A*, the gene encoding Nav1.7, is a cause of congenital insensitivity to pain, an extremely rare condition resulting in the inability to sense pain (Cox et al., 2006; Goldberg et al., 2007). Therefore, selective pharmacological inhibitors of Nav1.7 are being actively pursued as novel analgesics (Sun et al., 2014). The importance of Nav channels in the initiation and propagation of action potentials in nociceptive sensory neurons is undisputed, with local injection of non-selective Nav channel activators such as veratridine, ciguatoxin, grayanotoxin and aconitine causing spontaneous pain behaviours in mice, rats and humans (Focken et al., 2016; Gingras et al., 2014; Vetter et al., 2012b). These effects in turn can be reversed by Nav channel inhibitors, providing convenient *in vivo* target engagement models. For example, reversal of pain behaviours induced by local injection of aconitine was used to demonstrate *in vivo* target engagement of an aryl sulfonamide class of small molecules with Nav1.7 activity (Focken et al., 2016; Gingras et al., 2014). Given that activation of other Nav channel isoforms, including Nav1.1 or Nav1.6, is sufficient to elicit pain behaviours in rodents (Deuis et al., 2013; Osteen et al., 2016), the usefulness of this approach is particularly apparent for highly subtype-selective Nav channel activators. Recently, a Nav1.7-specific target-engagement model was developed based on intraplantar injection of the selective Nav1.7 activator OD1 (Deuis et al., 2016b). Similar to behaviours elicited by non-selective Nav channel activators, local administration of OD1 in mice causes rapid development of pain behaviours, including flinching and licking of the injected hind paw. Consistent with the high Nav1.7-selectivity of OD1, these behaviours were significantly attenuated in Nav1.7^{-/-} mice and by administration of selective Nav1.7 inhibitors GpTx-1 and PF-04856264 (Deuis et al., 2016b). OD1 is therefore a useful pharmacological tool to rapidly profile on-target activity of Nav1.7 inhibitors at sensory nerve endings *in vivo*.

5.3. Elucidating pain pathways

Five Nav channel subtypes (Nav1.1, 1.6, 1.7, 1.8, 1.9) are expressed on adult peripheral sensory neurons (Black et al., 1996; Fukuoka et al., 2008; Fukuoka and Noguchi, 2011). Due to a lack of subtype selective Nav channel inhibitors, determining the role of Nav channels in pain pathways has heavily relied on the knockout of Nav channel genes in mice. While this approach has been partially successful for Nav1.7, Nav1.8 and Nav1.9, global knockout of Nav1.1 and Nav1.6 is lethal as these isoforms are expressed in the central nervous system and Nav1.6 on the nodes of Ranvier in motor neurons (Caldwell et al., 2000; Gingras et al., 2014; Leo et al., 2010; Meisler et al., 2001; Whitaker et al., 2001; Yu et al., 2006). Instead, the role of Nav1.1 and Nav1.6 in specific pain pathways has been elucidated using selective Nav channel activator toxins.

Intraplantar injection of δ -theraphotoxin-Hm1a (5 μ M), a selective Nav1.1 activator with low nM EC₅₀ (38 nM), causes spontaneous pain behaviours and mechanical allodynia in mice, suggesting a role for Nav1.1 in mechanical pain (Osteen et al., 2016). However, at higher concentrations (>100 nM) Hm1a also inhibits K_V channels, including K_V2.1, K_V2.2, K_V4.1, K_V4.2 and K_V4.3 (Escoubas et al., 2002). K_V2.1 is a delayed rectifier channel that when open, causes outward current of potassium ions, allowing the cell membrane to repolarize after an action potential. It is expressed on peripheral sensory neurons and is important in the regulation of neuronal excitability; therefore, off-target activity at K_V2.1 could contribute to spontaneous pain behaviours observed after intraplantar injection of high concentrations of Hm1a (Kim et al., 2002; Tsantoulas et al., 2014). While Hm1a-induced spontaneous pain and mechanical allodynia was partially attenuated in conditional Nav1.1 knockout mice, basal mechanical sensitivity of this mouse line was not reported (Osteen et al., 2016). Therefore, further studies are required to confirm the role of Nav1.1 in mechanical pain and other pain modalities.

In peripheral sensory neurons, Nav1.6 is predominately expressed on the nodes of Ranvier, suggesting that it is crucial for the propagation of action potentials in myelinated fibers (Caldwell et al., 2000; Fukuoka et al., 2008). Indeed, intraplantar injection of the β -scorpion Cn2 (1–30 nM), a selective activator of Nav1.6 (EC₅₀ 39 nM), causes spontaneous pain behaviours, mechanical allodynia and enhances 4-aminopyridine-induced cold allodynia in mice (Deuis et al., 2013). A role for Nav1.6-expressing neurons in pathological cold pain was later confirmed using a combination of Nav channel knockout mice lines and selective Nav channel inhibitors in mouse models of oxaliplatin- and ciguatera-induced cold allodynia (Deuis et al., 2013; Minett et al., 2014; Vetter et al., 2012b; Zimmermann et al., 2013). While Nav1.6 is predominately expressed on A-fibres, it also expressed at comparatively low levels in C-fibers, where its role is unknown (Fukuoka et al., 2008).

5.4. Drug leads

Nav channel activator toxins may provide useful drug leads for the treatment of conditions caused by loss of Nav channel function. For example, selective Nav1.1 activators might find application as treatment for specific types of epilepsy, such as Dravet syndrome, which is associated with heterozygous loss-of-function mutations in *SCN1A*, the gene encoding Nav1.1 (Claes et al., 2001). Dravet syndrome is characterised by severe seizures that begin during infancy, followed by cognitive impairment, behavioural disorders and motor deficits that develop during early childhood. Potential therapeutics for Dravet syndrome would require exquisite selectivity for Nav1.1 over the other Nav subtypes, as non-selective activation of Nav channels would result in a limited therapeutic window, with high risk of seizures, cardiac arrhythmias and death.

However, given the poor prognosis and limited available treatments for Dravet syndrome, delivery of a Nav1.1 activator by an intrathecal pump that limits systemic exposure is a feasible option to improve the therapeutic window. Although less well-validated, Nav1.4 or Nav1.5 activators might hypothetically treat conditions such as congenital myasthenic syndrome, which is associated with severe muscle weakness; or cardiac arrhythmias and insufficiencies (Jurkat-Rott et al., 2010; Zimmer et al., 2014).

6. Conclusions

Nav channel activators, produced by bacteria, dinoflagellates, algae and many venomous animals including scorpions, sea anemone, wasps, spiders, cone snails and snakes, are a structurally and functionally diverse group of compounds that interact with multiple sites on Nav channels to enhance function. In practice, the effects elicited by Nav channel activators are as diverse as their origins, and include enhanced activation, decreased inhibition or altered conductance. However, it is becoming increasingly clear that this remains a simplified view of the pharmacology of Nav channel activators. For example, enhanced activation can include effects on the voltage-dependence of activation or changes in channel opening kinetics. Similarly, effects on inactivation can manifest as altered voltage- or time-dependence of inactivation; effects on fast or slow inactivation; altered recovery from inactivation, incomplete inactivation or emergence of resurgent or leak currents. Moreover, the assumption that biophysical effects on channel gating are similar across different isoforms clearly does not hold true. In combination with different binding affinities for individual isoforms, we are just beginning to understand the pharmacological complexity of Nav channel activators. Accordingly, systematic pharmacological characterisation of these compounds, in light of better molecular tools and improved crystal structures available to us now and in future, will undoubtedly provide additional insight into the therapeutic and practical applications for Nav channel activators.

Conflict of interest

The authors declare no conflicts of interest.

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